

Task prioritization and distributed deep reinforcement learning for healthcare management in Cloud-Edge environments

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ABSTRACT

Task offloading-based healthcare management in a Cloud-Edge environment emerges as a transformative approach that dynamically distributes computational tasks between Internet of Things devices, edge servers, and centralized cloud infrastructures. In this research, a Task-Prioritized Distributed Stacked Deep Reinforcement Learning strategy is proposed for task offloading-based healthcare management in a Cloud-Edge Environment. The physiological data are gathered continuously from the Internet of Things-enabled medical devices in the device layer, which are transferred into the edge layer for the operations of preprocessing for enhancing the quality of the data. The Hybrid Temporal-Contextual Fusion module is implemented by incorporating the Transformer-based attention and Bidirectional Long Short-Term Memory networks to capture the temporal dependencies and contextual relationships in physiological signals. A Health Severity Index is analyzed for assessing the risk levels of the patient and allows dynamic task prioritization for monitoring the healthcare tasks based on the extracted features. An Uncertainty-Aware Priority-Driven Distributed Stacked Deep Reinforcement Learning model at the cloud layer resolves the decisions of optimal task offloading between the device, edge, and cloud resources. A Stacked Convolutional Neural Network is utilized for learning the system state's compact representation, and the Q-values are evaluated for offloading actions. An uncertainty-aware learning mechanism and a multi-objective reward function are integrated for enhancing the reliability of the decision. The proposed technique is compared to previous mechanisms, attaining a minimum Latency of 170 ms, and a higher resource utilization rate of 84 % conforming that the proposed approach substantially enhances energy efficiency and overall performance in task offloading-based healthcare management within Cloud-Edge environments.

1. Introduction

In the modern era, healthcare management has become increasingly reliant on digital technologies to ensure the timely and effective delivery of medical services [1]. With the exponential growth in the number of connected medical devices, including Electronic Health Records (EHRs) and real-time patient monitoring systems, the volume of healthcare data being generated has surged dramatically [2]. This rapid digitalization demands robust computational infrastructures capable of handling, processing, and analyzing healthcare data efficiently. Traditional cloud-based solutions often face latency and bandwidth issues, while offering vast storage and computing capabilities, particularly when real-time decision-making is required [3]. This has led to the emergence of cloud-edge computing paradigms that aim to balance the load between cloud and edge devices, which offers both the scalability of cloud systems and the low-latency responsiveness of edge nodes.

In this context, task offloading becomes a crucial function, where healthcare tasks ranging from data collection and pre-processing to advanced analytics and diagnostics are intelligently distributed between edge and cloud environments [4]. Effective task offloading ensures optimal use of computational resources, reduced response time, and improved system efficiency. However, all the tasks in healthcare applications do not have equal importance; some are time-critical and demand immediate attention, such as emergency response or real-time vital sign monitoring, while others, like routine data backups or statistical analyses, tolerate delays [5]. Therefore, a task prioritization model becomes essential to identify and process high-priority healthcare tasks, which enhances the overall quality of service and patient care.

Reinforcement Learning (RL) has emerged as a powerful tool for optimizing healthcare management by enabling intelligent decision-making through continuous interaction with dynamic environments [6]. In healthcare systems, RL is employed to personalize treatment

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plans, optimize hospital resource allocation, and improve patient monitoring by learning optimal policies based on real-time data and outcomes [7]. For instance, RL agents adaptively manage task offloading in cloud-edge environments by learning from historical and contextual information to minimize latency and maximize resource efficiency [8]. Unlike traditional rule-based systems, RL offers the advantage of self-improvement over time, which makes it particularly suitable for complex, data-intensive healthcare scenarios where decisions must adapt to changing patient conditions, network states, and system loads [9]. This capability positions RL as a promising approach for creating adaptive, responsive, and patient-centric healthcare solutions.

Despite this research, existing offloading strategies often process tasks homogeneously, neglecting the critical aspect of task prioritization. These methods lead to increased delays for time-sensitive applications, inefficient resource utilization, and suboptimal patient outcomes [10]. Moreover, many current techniques rely on static decision-making processes that do not adapt well to the dynamic and heterogeneous nature of healthcare environments. Previous RL-based task offloading techniques face limitations such as slow convergence, high computational overhead, and inadequate adaptability to dynamic healthcare settings. These limitations necessitate a more robust and adaptive strategy for task offloading-based healthcare management. To address these limitations, the proposed work introduces an optimized task-offloading approach tailored for healthcare management within a cloud-edge ecosystem. This research proposes a novel Task-Prioritized Distributed Stacked Deep Reinforcement Learning (DRL) (DSDRL) (TP-DSDRL) approach for intelligent task offloading based on Healthcare Management in Cloud-Edge Environment. The proposed framework aims to enhance resource utilization efficiency and minimize energy consumption, latency, and computational cost.

1.1. Novelty

The novelty of the proposed TP-DSDRL method in terms of task offloading of healthcare information in the cloud-edge computing environments is presented in this section.

- A TP-DSDRL framework is proposed for offloading the healthcare task intelligently in cloud-edge computing environments. In this method, the network conditions, availability of system resources, and patient risk are combinably considered to produce the decisions of offloading adaptively.
- A Hybrid Temporal-Contextual Fusion (HTCF) method is implemented through the incorporation of Bidirectional Long Short-Term Memory (BiLSTM) along with the Transformer-based attention mechanism. With this integration, the contextual relationships and temporal dependency in physiological signals are captured, resulting in enhanced healthcare accuracy in analyzing severity.
- A dynamic task prioritization mechanism is adopted for the healthcare task classification according to the clinical urgency. The physiological indicators, including the status of patient condition, ECG abnormalities, and heart rate (HR), are employed for the priority score computation, allowing to classify the task into low, medium, high, and emergency priority levels.
- For enhancing the decision-making reliability, an Uncertainty-Aware Priority-Driven DSDRL (UAP-DSDRL) is integrated to analyze the predicted Q values confidence through variance based analysis. Based on this mechanism, the increased risk of the actions of offloading is prevented during lower prediction confidence.
- A Stacked Convolutional Neural Network (SCNN) is incorporated into the architecture of reinforcement learning for processing the state information at high dimensional system, such as availability of cloud resources, edge workload, network bandwidth, device energy level, and task priority.

1.2. Contribution

These contributions to task offloading-based healthcare management in Cloud Edge Platform are as follows:

- A TP-DSDRL framework is proposed for the intelligent task offloading in a healthcare system in cloud-edge computing environments. The network conditions, system resource availability, and patient risks are combinably considered for producing the offloading decisions adaptively.
- An HTCF module is developed by incorporating the BiLSTM along with the Transformer-based attention mechanism, in which the contextual relationships and temporal dependencies are captured in physiological signals, enhancing the accuracy of analyzing the healthcare severity.
- A dynamic task prioritization mechanism is implemented for healthcare tasks classification based on clinical urgency. The physiological indicators, including patient condition status, ECG abnormalities, and HR, are utilized for priority score computation, allowing the classification of tasks into priority levels of emergency, high, medium, and low.
- A UAP-DSDRL method is integrated to enhance the decision-making reliability and for assessing the predicted Q-values confidence through the variance-based evaluation, thereby preventing high-risk offloading actions when there is a reduced prediction confidence.
- A SCNN is incorporated in the architecture of reinforcement learning for effectively executing the high-dimensional system state information, such as device energy level, task priority, edge workload, network bandwidth, and availability of cloud resources.

This research paper is structured as follows: [Section 2](#) presents a comprehensive literature review, and [Section 3](#) outlines the system model and formulates the problem statement. [Section 4](#) details the proposed TP-DSDRL strategy, while [Section 5](#) discusses the experimental setup and analyzes the results. [Section 6](#) concludes the dissertation and outlines potential future research directions.

2. Literature review

Recent studies have explored a variety of task offloading approaches for healthcare management within cloud-edge platforms. The related work on these previous techniques is discussed below.

2.1. Literature related to task offloading for healthcare management in cloud-edge platform

Bhasker et al., [11] presented an efficient augmented moth-flame optimization (EA-MFO) approach for managing task offloading in the Internet of Medical Things (IoMT) cloud, specifically designed for healthcare applications. EA-MFO used a chaos-based logistic map for initializing moth positions to ensure high population diversity and better exploration of the search space. It optimizes the power consumption and makespan with varying cloud resources. Jameil and Al-Raweshidy [12] suggested a Digital Twin Healthcare enhanced asynchronous team-based multi-agent proximal policy optimization (DTH-ATB-MAPPO) for medical task offloading. The architecture is compatible with various communication protocols and relies on partial and binary offloading for smooth data transfer. MAPPO learns and adapts to the environment dynamically to enhance the quality and resource utilization efficiency for decision-making.

Mondal et al., [13] presented a dynamic energy-efficient offloading strategy (DEEO) for task offloading in a mobile edge-cloud computing system. In DEEO, a mobile device can offload heavy and secure image transfer tasks to the nearest edge server/fog access point to reduce energy consumption while ensuring the safety and integrity of health data. A Dynamic Priority-Based Task Scheduling and Adaptive Resource

Allocation (DPTARA) was designed by Anand and Karthikeyan [14] for healthcare task on real time based on system load and urgency. The healthcare task prioritization was employed by this method to reduce the latency and for resource allocation to the end devices. The resource utilization was optimized, and the load balancing over the nodes of the edges was attained to improve the performance of the system.

An online Offloading based on the DRL was presented by Su et al., [15], which incorporates a deep neural network and a weight-sensitive experience replay mechanism for resolving the real-time task execution of trade off of the latency and energy consumption of the system. The DNN was utilized for reducing the decision of task offloading; meanwhile, in the experience memory, the optimal solution was deposited and weighted the samples based on the training effect. He et al., [16] suggested a mobile edge computing (MEC)-enabled system for constructing an intelligent multitasking processing system for data offloading in smart healthcare. The task-aware service offloading algorithm is developed through deep deterministic policy gradient (DDPG), which provides adaptability in terms of various services via end-edge-cloud synergy.

Furthermore, Saranya et al., [17] suggested a DRL process together with a Lion Optimization Algorithm (DRL LOA). This helped to enhance the quality of decisions made in a cloud-edge environment. This mechanism utilized a Graph Convolutional Network (GCN) for combining this information into traffic attributes to create a node vector, thus enhancing the efficiency of categorization for medical data in a cloud edge setup. A Priority-Based Dynamic Load Balancing (PBDLB) algorithm was designed by Shrawankar et al., [18] for leveraging the criticality of the task for effective distribution over the nodes of the fog. The categorization of tasks into high-priority and low-priority was performed to ensure that the immediate processing was received through the time-sensitive operations. The dynamic resource weighting was integrated to reflect the workload in the real world situations.

Choppara and Lokesh [19] suggested a DRL strategy in load balancing and optimizing tasks in fog and cloud computing in critical healthcare. This mechanism utilizes the prioritized tasks strategy depending on heart rate and ECG parameters to help in the efficient handling of critical tasks in healthcare. The platform streamlined fog node resource utilization and eliminated the delay in critical healthcare data. A hybrid computing model was demonstrated by Khan et al., [20] in a healthcare system that utilized genetic algorithm optimization and Graph Neural Networks to identify workloads and networks in Edge and Cloud layers. This mechanism led to power savings and decreased latency.

Motlagh et al., [21] have demonstrated a detailed review of the predictions of personality traits using AI. They have also mentioned the requirements for explainable AI and the benefits of using a combination of ML and DL in improving intelligent decision-making systems. Asif et al., [22] have done an exhaustive survey on the applications of ML in early disease detection in many medical disciplines like cancer, diabetes, and COVID-19.

A Traffic-aware lightweight Hierarchical Offloading framework towards Adaptive Slicing-enabled Space-Air-Ground Integrated Networks (SAGIN) was developed by Chen et al., [23], in which the SAGIN is separated into Communication Access Platforms (CAPs) and Computation Offloading Platforms (COPs). A self-attention-based prediction model was implemented accurately to allow adjustments in the slice resource. An improved reinforcement learning method was developed based on clipping of proximal along with dynamic confidence intervals for optimal offloading. A personalized Federated deep Reinforcement learning based computation Offloading and resource Allocation method (PFR-OA) was developed by Chen et al., [24] in which personalized demands are considered in smart communities during the proper policy generation of resource allocation and computation offloading. A proximal term was implemented to enhance the optimal loss function of local Q-values in reinforcement learning.

A Multi-agent Collaborative method for Workflow Task offloading in

end-edge-cloud environments with the Actor-Critic algorithm named MCWT-AC was presented by Xiao et al., [25], where every MD was initially formed as an agent, and the decision of offloading was made independently according to the local information. The workflow of every MD of the task offloading decision model was acquired using the Actor-Critic algorithm. An efficient workflow task offloading plant was implemented at runtime using multi-agent collaboration. An Internet of Healthcare Things (IoHT) framework was designed by Van Anh and Nguyen [26], incorporating an M/M/C/K priority queue model (M: Markovian arrival/service rates, C: servers, K: capacity) along with an edge-fog-cloud system of a three-tier architecture. A dynamic priority assignment method was designed for the rapid execution of critical events using the real-time data of the patient; moreover, an adaptive resource allocation method was exploited for performance optimization over different workloads.

2.2. Limitations and research gap

The existing work perform important role in increasing computational efficiency and decreasing latency due to task offloading in cloud-edge computing, but it still contains some limitations. The majority of existing methods deal with the tasks similarly, without focusing on the task priority or application emergency, which causes complex conditions in healthcare where delay-sensitive needs instant processing. Moreover, some of the task offloading methods depend on static system parameters that include device energy, computational capacity, and network bandwidth, but fail to adopt the temporal dynamics and contextual relationships in the real-world data streams. Although DRL is commonly accepted for adaptive offloading decisions, most of the conventional method has absence of mechanisms to calculate the uncertainty of predicted actions, which causes unpredictable decisions across highly dynamic network environments. Another important limitation lies in the insufficient representation of high-dimensional system states. In many existing models, the ability to extract complex features associated with workload fluctuations, resource availability, and network congestion remains limited. As a result, these approaches often fail to fully capture the intricate relationships that influence offloading decisions. Therefore, a clear research gap exists in the development of intelligent, distributed, and priority-aware task offloading frameworks capable of effectively modeling temporal and contextual information. Such frameworks should also incorporate uncertainty-aware decision mechanisms while efficiently handling high-dimensional system states to ensure reliable performance for latency-sensitive applications in cloud-edge computing environments. To address these issues, this study proposes a TP-DSDRL framework. The proposed approach integrates priority-based decision making, temporal and contextual learning, and uncertainty-aware reinforcement learning to enable efficient task offloading.

3. System model

The proposed TP-DSDRL method's system architecture and the mathematical formulation, which are employed for task offloading efficiently in cloud-edge healthcare environments, are presented in this section. The main aim of this system is to allow the healthcare devices based on IoT to generate the task offloading decisions optimally, which minimizes latency, reduces consumption of energy, and the healthcare monitoring is managed reliably. The determination of the most appropriate location (cloud, edge, or device) for execution of the tasks of healthcare is focused on the problem formulation during the system constraints satisfaction relevant to cost, energy, and delay in processing.

The architecture of healthcare monitoring is displayed in Fig. 1. Three hierarchical layers, including the Device layer, the Edge layer, and the Cloud layer, are contained in this architecture.

The heart-rate sensors, ECG monitors, wearable sensors, and the various IoT-activated medical devices accountable for the physiological

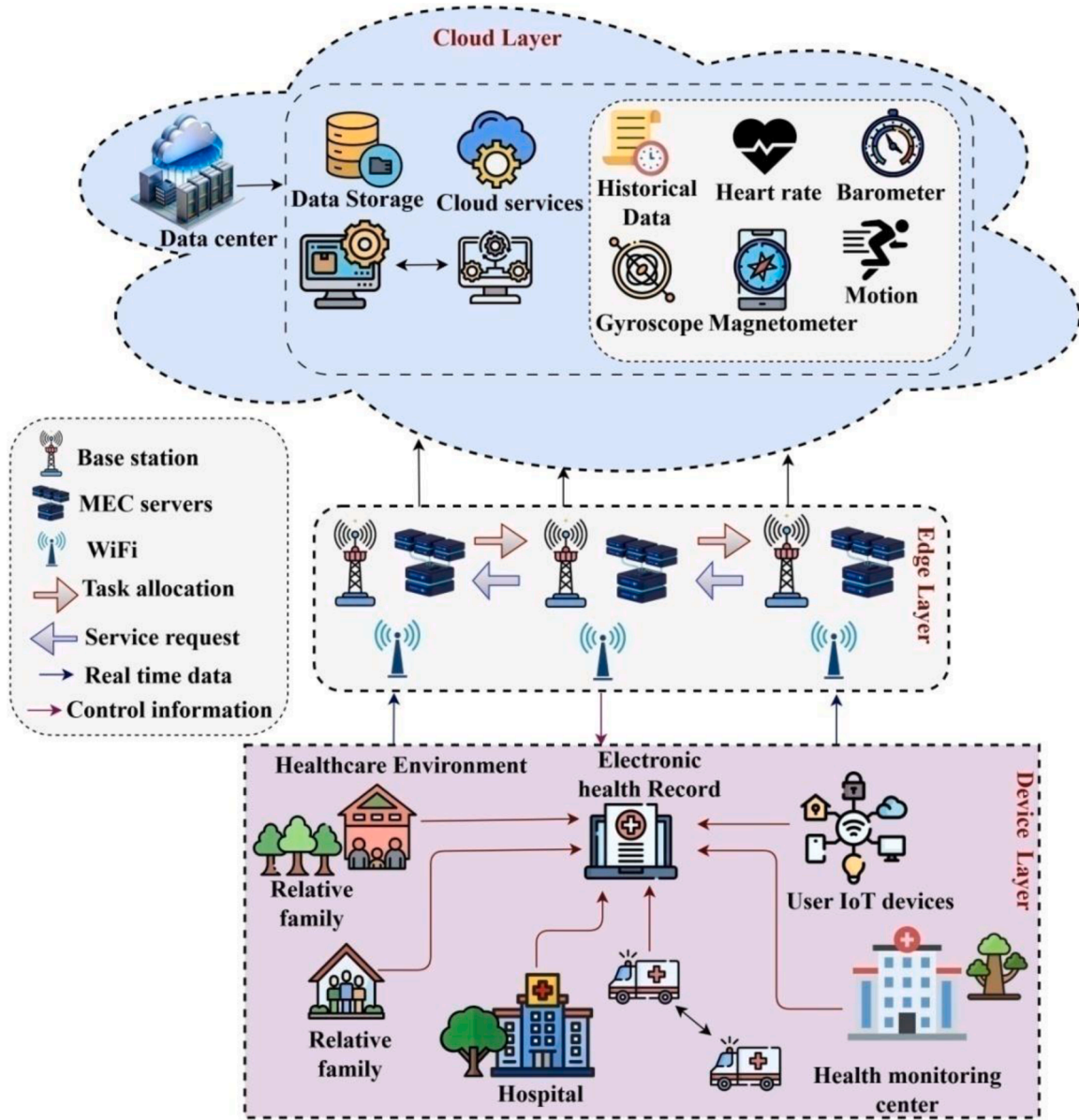


Fig. 1. Architecture of healthcare monitoring.

data collection from patients are contained in the device layer. The healthcare parameters are monitored continuously through these devices, resulting in computational task production for processing. External computing support is required because of reduced energy constraints and computational resources.

The edge servers are contained in the edge layer developed near network gateways and healthcare facilities. The intermediate computing abilities are provided to execute healthcare tasks that are sensitive to latency, including emergency alerts or anomaly detection. The real-time responsiveness is enhanced, and the communication delay is minimized significantly in the system through the partial computation performance at the edge.

The computing resources that are performing extensively for long term storage, large-scale processing of data, and complicated analytics are offered in the cloud layer. From various edge nodes, the information is aggregated through the cloud server, resulting in the global optimization processing through the proposed TP-DSDRL method, assessing the policy of optimal task offloading based on the conditions of the system.

Effective collaboration among the sensing devices and allocated

computing resources is allowed through the multi-layer architecture, assisting continuous monitoring of patients, intelligent management of healthcare, and emergency detection.

3.1. Problem formulation

The assessment of an optimal decision policy that reduces latency, energy consumption, and cost of execution while assuring reliable processing of healthcare data is the main objective of the task offloading problem.

Let IH and ES represent the healthcare devices based on IoT and edge servers in the system. A set of computational tasks is produced by every device, which is locally executed or offloaded to a cloud or edge server.

The representation of the task produced through their device is formulated as follows:

$$\tau_i = (DS_i, CC_i) \quad (1)$$

From the above equation, the input size of data of the task is signified as DS_i and the essential Central Processing Unit (CPU) cycles for execution are indicated as CC_i .

• Energy Consumption Model

When the tasks are locally processed or transformed to remote computing nodes, the consumption of energy is consumed. The definition for the consumption of energy for the processing of a local task at time is:

$$EC_i^{lo} = \kappa CC_i f_i^2 \quad (2)$$

In the above equation, the CPU frequency for the device is represented by f_i , and the processor's efficient switched capacitance coefficient is denoted as κ .

The energy essential for the computational task processing on the resource-limited IoT devices is considered through this model.

The Dynamic Voltage and Frequency Scaling (DVFS) is often utilized by the processor in practical edge and IoT computing systems to adapt the frequency of the CPU based on the conditions of the workload. The dynamic variations of processor frequency and voltage in DVFS result in the consumption of energy that is nearly proportional to the operating frequency cube. The maximum usage of energy is led by the increased frequency execution, and the consumption of power at greater execution time cost is minimized while at reduced frequency. In the latest edge-enabled healthcare systems, a more realistic energy consumption expression is offered using DVFS.

• Latency Model

The entire latency of processing a task consists of computation delay and transmission delay. Three processing modes are utilized for the execution of every task in the proposed TP-DSDRL model, and they are local processing, edge offloading, or cloud offloading.

The following expresses the entire latency execution for T_i task:

$$LT_i = LT_i^{lo} + LT_i^{eg} + LT_i^{cd} \quad (3)$$

From the above equation, the local computation delay is signified as LT_i^{lo} , the latency execution at edge layer is indicated as LT_i^{eg} , the delay related with data transmission and cloud processing is represented as LT_i^{cd} . The computing node's resource and storage constraints need to be satisfied during the execution of the task.

• Execution Cost Model

The total cost of task execution is affected by the energy consumption and latency. The cost of local execution for the device is expressed as:

$$CS_i^{lo} = \alpha LT_i^{lo} + \beta EC_i^{lo} \quad (4)$$

In the above equation, the weighting factors illustrating the significance of energy consumption and latency are indicated by α and β .

The definition of the entire cost of offloading is expressed in the following;

$$CC_i = \alpha LT_i + \beta EC_i \quad (5)$$

From Eq. (5), the overall energy consumption and latency related to the task execution are denoted as EC_i and LT_i .

A penalty term is implemented for the task failure representation when the task is not executed because of inadequate communication failure among the edge server and the IoT device or computational resources.

The most significant role is functioned by the penalty function in enhancing the robustness of the system. It functions while the constraints of task execution are violated because of network outages, edge server failures, or inadequate computing resources. The penalty term maximizes the cost of execution when these failures occur, which discourages the policy of offloading from unreliable computing node selection. This function permits the proposed TP-DSDRL method to preserve healthcare service delivery reliably and prevent unstable

processing environments.

• Fitness Function

The proposed task offloading function's performance is assessed through a fitness function that includes execution cost, latency, and energy consumption into an individual objective measure.

$$FS = \omega_1 EC + \omega_2 LT + \omega_3 CS \quad (6)$$

From the above equation, the entire system execution cost, latency, and energy consumption are signified as CC , LT and EC . The weighting parameters are the ω_1 , ω_2 and ω_3 coefficients that manage the relative significance of every factor.

The reduction of fitness function during the timely task execution and reliable healthcare monitoring maintenance is the main goal of the risk-aware TP-DSDRL method.

4. Proposed methodology

A TP-DSDRL framework is proposed for intelligent offloading of healthcare tasks in cloud edge computing environments. Three hierarchical layer including the device layer, edge layer, and cloud layer is contained in this method. The blood pressure, oxygen saturation, heart rate, and the ECG signals are the physiological data that are gathered continuously from the medical IoT-based devices in the device layer. To the edge layer, the gathered data are passed for the execution of pre-processing operations such as noise removal, normalization, and missing value handling to enhance the data quality. These preprocessed data are considered through the HTCF module, which incorporates Transformer-based attention and BiLSTM networks for capturing temporal dependency and contextual relationships in physiological signals. A Healthy Severity Index (HSI) is assessed based on the features extracted to analyze the risk level of the patient and allow the prioritization of tasks dynamically for monitoring the task based on healthcare. A comprehensive state representation is constructed by the system that contains the resource availability, network conditions, device energy level, and the severity information of healthcare. In the cloud layer, optimal task offloading decisions between the cloud resources, edge, and device are determined using the UAP-DSDRL model. A SCNN is utilized for learning the system state's compact representation, and the Q-values are evaluated for offloading actions. An uncertainty-aware learning mechanism and a multi-objective reward function are integrated to enhance the reliability of decisions during the reduction of operational cost, energy consumption, and latency. An adaptive offloading strategy is learned by the proposed framework using continuous interaction along with the cloud-edge environment, which assures timely execution and effective utilization of resources for important healthcare tasks. Fig. 2 illustrates the structure of the proposed TP-DSDRL method.

4.1. Data acquisition

At the device layer, the processing of data acquisition based on healthcare is performed in the proposed TP-DSDRL method. This contains various medical IoT devices, and it is allowed to monitor the physiological conditions of the patient continuously. These devices are blood pressure monitors, pulse oximeters, heart rate sensors, ECG monitors, wearable biosensors, and various smart health tracking systems. Continuous physiological data streams, including body temperature, blood pressure (BP), oxygen saturation (SpO2), heart rate (HR), and CG signals, are necessary for monitoring real-time health and to identify abnormal conditions early.

For the edge layer, the gathered physiological signals are transformed using the wireless communication technologies, including cellular networks, Bluetooth Low Energy (BLE), or WiFi. The data acquisition process is developed for managing the streaming of data continuously during the reliability maintenance and reducing

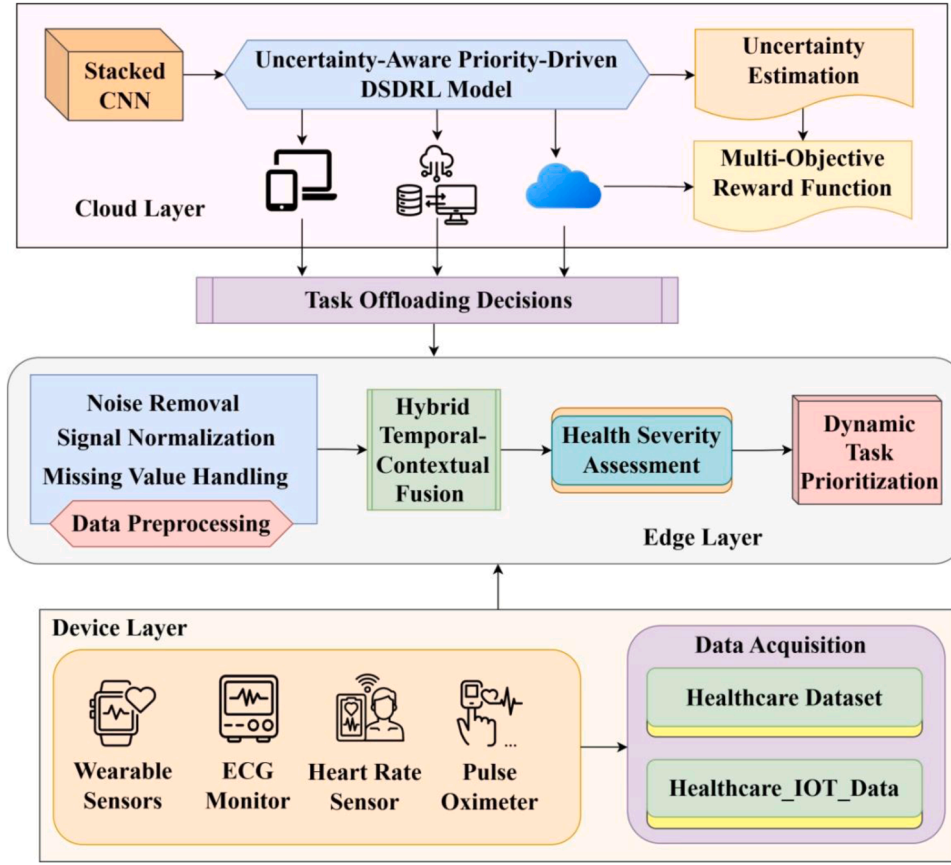


Fig. 2. Structure of the proposed TP-DSDRL method.

communication latency, even the sensor data of healthcare is often time-dependent and heterogeneous. Along with the physiological dimensions, the contextual information, including environmental conditions, patient activity level, and device status are also gathered to offer a more extensive representation of the health state of the patient.

4.2. Data preprocessing

The incoming raw data based on healthcare encounter systematic preprocessing in the edge layer to improve the quality of the signal, develop reliable inputs for consequent task prioritization, temporal feature learning, and assure data consistency. Continuously, several preprocessing operations are executed based on the noise-prone and heterogeneous characteristics of physiological signals based on IoT.

Noise Removal: From monitoring devices and wearable devices, the physiological signals are gathered, and they are mostly infected using sensor instability, environmental interference, and motion artifacts. To solve these problems, suitable filtering methods are utilized to eliminate the baseline drift and noise at high frequency while maintaining the substantial signal patterns clinically. The distorted representation of the signal is prevented, which adversely affects the learning performance and estimation of downstream severity.

Signal Normalization: The features are scaled into a uniform range in normalization since different numerical units and ranges are exhibited using the acquired healthcare parameters. The numerical stability of the model is assured, and the bias is prevented over attributes at high magnitude while training the model. The BiLSTM-Transformer model's convergence behavior is enhanced through proper normalization, resulting in an improvement in the entire efficiency of learning.

Missing Value Handling: Missing samples are contained in the healthcare time-series data because of device malfunction, intermittent connectivity, or packet loss. Through statistical imputation or

interpolation methods, the missing values are handled to avoid information loss and to preserve temporal continuity, and the input sequences' structural completeness is assured prior to temporal modeling.

4.3. Temporal and contextual feature learning

The preprocessed physiological series are passed to an HTCF module to withdraw the severity-aware and discriminative representations of the feature. The self-attention-based contextual modeling is integrated along with the bidirectional temporal dependency learning to capture the long-range sequential patterns and inter-parameter correlations inherent in time series data on healthcare. The multi-head self-attention mechanism is utilized to model the contextual dependency between the physiological parameters, thereby allowing the clinically substantial indicator's adaptive weighting. Following, the attention refined features are processed using the BiLSTM for backward and forward temporal dependency learning, assuring the anomaly trends and dynamic health fluctuation modeling. Robust embedding of health state is produced by the HTCF module, which offers the basis for priority-driven task offloading and accurate severity evaluation. Design of the HTCF method is shown in Fig. 3.

The temporal transformer encoder phase is utilized by this HTCF, resulting in preprocessed physiological sequences processing, which is then transformed into a group of substantial feature vectors. This comprises various encoder layers in which each includes two sub layers that are multi-head self-attention and a feed-forward network that are fully connected in vocationally. For every two sub-layers, the residual connections are employed by the transformer to improve the information flow and alleviate the problem of vanishing gradient. These sub-layers are integrated using the layer normalization, and the formulation of the output of every sub-layer is $\text{LayerNorm}(a + \text{sublyr}(a))$ where the operation executes, using the sub-layer itself is signified by $\text{sublyr}(a)$. An

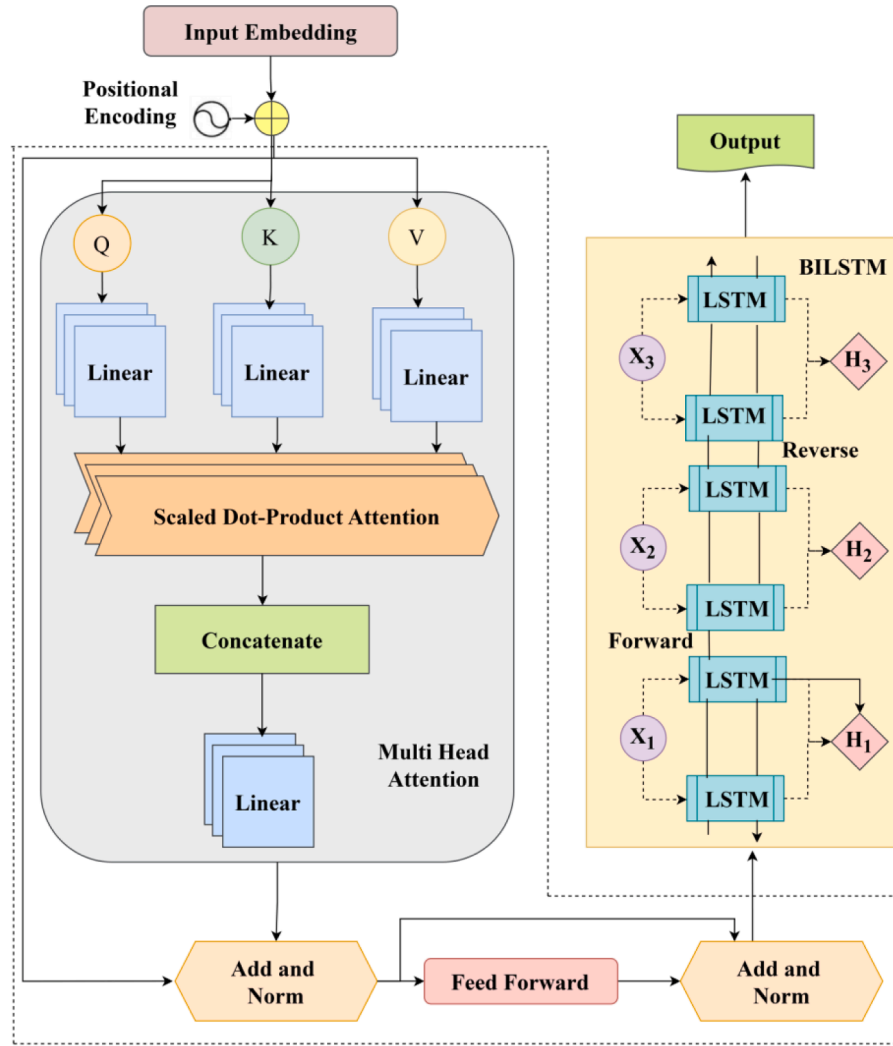


Fig. 3. Design of the HTCF method.

output along with D_{mdl} dimensionality is produced by entire sub-layers, such as embedding layers, to allow smooth incorporation of these residual connections.

The physiological segment sequences that are clinically appropriate are focused on by the model in the attention mechanism. These functions are performed through a set of key-value pairs and a query vector. The contemporary contextual focus of the model in the health signal is represented using the query; meanwhile, various physiological measurements' temporal segments, including oxygen saturation, QT interval, and heart rate, are expressed using the key-value pairs. The important health indicators participating to the estimation of severity are emphasized dynamically using the model.

The scaled dot-product method, named self attention computation method, is the main method employed for attention relationship computation. The keys included into K matrix, and the queries included in the Q matrix are the input, which consists of D_k dimension and the values are loaded into V the matrix, which consists of D_v dimensions. The following equation is utilized for the computation of the output matrix:

$$Attn(Q, K, V) = \text{softmax}\left(\frac{QK}{\sqrt{D_k}}\right)V \quad (7)$$

The h linear projections are utilized by the multi-head attention instead of using an individual attention function queries, values, and keys D_{mdl} dimension. Through these projections, the queries, keys, and values are transformed into dimensions of D_q , D_k , and D_v in which every

projection is independently learned. In every projection version, the attention function is executed simultaneously, generating output values at D_v the dimension. The information from varied subspace representations and several positions is determined efficiently at once by the model when initiating multi-head attention.

$$MH(Q, K, V) = \text{Concat}(hd_1, hd_1, \dots, hd_k)WP^0 \quad (8)$$

where $hd_i = Attn(QWP_i^Q, KWP_i^K, VWP_i^V)$

From the above equation, the learned projection matrices are signified as WP_i^Q , WP_i^K and WP_i^V . Varied contextual relationships among various temporal positions and physiological parameters are captured collectively by the model through the formulation of multi-head attention.

The extracted embedded features of healthcare time series after preprocessing are received by the transformer encoder in the HTCF module. The inter-parameter correlations and long-range dependencies across the overall sequence are modeled by the encoder in self attention mechanism. In monitoring healthcare, the ability of this function is essential, in which the physiological abnormalities mostly show subtle or delayed temporal interactions. A holistic dynamic health pattern representation is gained using the model through attending to a similar context over the entire sequence.

A point-wise feed-forward network is utilized after the self-attention operation for improving the capacity of representation and implement non linearity. Two linear transformations are contained in the feed-

forward network, which are segregated through an activation function, allowing complicated nonlinear interactions modeling between the physiological variables. The contextual embeddings are refined through these transformations and generates discriminative representation of features appropriate for dynamic task prioritization and severity assessment.

In the HTCF module, the produced contextual features from the transformer encoder are passed into the BiLSTM layer. The attention-enhanced representations are refined using the BiLSTM network, where the bidirectional temporal dependency is modeled, which is available in physiological time series data. The backward and forward sequences are processed through the BiLSTM, allowing extensive analysis of health pattern evolution. In signals, the progressive trends and historical dependency, including oxygen saturation, variability of heart rate, and ECG, are captured through the forward LSTM, while the future contextual relationships related to every time step are learned simultaneously by the backward LSTM. The capability of the model is improved through these dual-directional processing to recognize physiological responses that are delayed and subtle temporal variations, which are significant for accurate assessment of health severity.

The computation of backward and forward hidden states for an a_t input sequence is formulated as follows:

$$\overrightarrow{hs}_t = LSTM_{fd}(a_t, \overrightarrow{hs}_{t-1}) \quad (9)$$

$$\overleftarrow{hs}_t = LSTM_{bd}(a_t, \overleftarrow{hs}_{t+1}) \quad (10)$$

Both the hidden states are concatenated, and it is utilized for obtaining the final bidirectional representation, which is formulated as:

$$hs_t^{Bid} = [\overrightarrow{hs}_t; \overleftarrow{hs}_t] \quad (11)$$

The extensive past and future contextual information is captured by the fused temporal representation, and the selective severity-aware embeddings are generated. The final temporal contextual feature vector is offered as the BiLSTM layer output, which is employed for the dynamic task prioritization and health severity assessment in the proposed TP-DSDRL method.

4.4. Health severity assessment

A quantitative Health Severity Index (HSI) is measured after the temporal-contextual representation extraction using the HTCF method for assessing critical conditions in clinics for every monitoring task. In this phase, the multimodal physiological measurements are transformed into a combined severity score, which guides risk-aware offloading decisions and dynamic task prioritization in the proposed method.

The clinically essential physiological indicators, such as systolic blood pressure (SBP), oxygen saturation (SpO₂), corrected QT interval (QTc), and heart rate (HR), are integrated through the severity assessment. Based on their developed combination with hemodynamic abnormalities, hypoxemia, and cardiac instability in real-time monitoring of patient systems, these parameters are chosen.

4.4.1. QT interval-based severity modeling

The QT interval is chosen between the derived parameters as the main severity indicator because of its robust clinical association with ventricular arrhythmias and unexpected cardiac death. Ventricular repolarization electrical instability is reflected by QT prolongation, which is mostly identified as a forecaster of malignant arrhythmic circumstances. In wearable monitoring environments, the QT-based measurements are stable and low susceptible to the artifacts of transient motion. The health severity index major component is adopted by the corrected QT interval in the proposed method.

$$QT_c = \frac{QT_{invl}}{\sqrt{RR_{invl}}} \quad (12)$$

In the above equation, the measure QT interval is signified as QT_{invl} and the R-R interval is indicated as RR_{invl} . The maximized electrical instability is indicated by the prolonged QTc values, thereby allocating increased severity contributions.

4.4.2. Clinical threshold-based risk functions

The developed clinical thresholds are employed for the assessment of abnormalities in heart rate:

$$\begin{aligned} HeartRate &> 120 \text{ bpm} \\ HeartRate &< 60 \text{ bpm} \end{aligned} \quad (13)$$

Maximum clinical risk is indicated by the minimized abnormal blood pressure and minimized SpO₂ (< 95 %). The conversion of every physiological parameter into a normalized risk function is performed, and it is expressed as:

$$F_i = \frac{a_i - a_i^{nr}}{a_i^{mx} - a_i^{mn}} \quad (14)$$

From the above equation, the observed value is signified as a_i and the normal reference value is signified as a_i^{nr} .

4.4.3. Health severity index formulation

The evaluation of the Health Severity Index is based on a weighted sum of normalized risk components

$$HSI = \omega_1 F_{HR} + \omega_2 F_{QTc} + \omega_3 F_{SpO_2} + \omega_4 F_{BP} \quad (15)$$

In this equation, ω_i is denoted as a weighting coefficient satisfying:

$$\omega_1 + \omega_2 + \omega_3 + \omega_4 = 1 \quad (16)$$

These weighting coefficients are adapted based on system feedback for significant responsiveness to each parameter.

4.4.4. Severity classification

According to the computed HSI, the patient conditions are divided into priority levels:

$$\rho = \begin{cases} \text{High,} & HSI \geq \theta_1 \\ \text{Medium,} & \theta_2 \leq HSI < \theta_1 \\ \text{Low,} & HSI < \theta_2 \end{cases} \quad (17)$$

From this equation, ρ indicates priority, θ_1 and θ_2 are denoted as predefined thresholds.

The division of severity provides efficient and risk-aware task prioritization in the proposed TP-DSDRL model.

4.5. Dynamic task prioritization

The highest numbers of physiological monitoring programs are consistently produced from wearable sensors and medical devices in IoT-provided healthcare monitoring systems. The physiological monitoring programs are crucially varied in the field of clinical urgency, risk level, and processing deadlines. Accessing all tasks with the same level of priority can causes difficult in critical medical situations and time delay for processing. Moreover, the proposed TP-DSDRL model integrates a dynamic task prioritization mechanism for classifying healthcare monitoring tasks according to clinical severity and urgency.

A wearable or medical sensing device producing i^{th} a monitoring task is indicated as τ_i . All tasks are related to physiological indicators and contextual parameters from the feature extraction. The estimation of the priority score process uses prioritization, which identifies the urgency level of the task and manages the offloading decisions in the proposed TP-DSRL technique.

The priority score can be formulated as:

$$\rho_i = \alpha HSI_i + \beta \frac{1}{d_i} + \chi r_i - \lambda u_i \quad (18)$$

In this equation, $\alpha, \beta, \chi, \lambda$ are the weighting coefficients satisfy, HSI_i indicate the health Severity Index of the patient i , d_i denote the task, r_i is the clinical risk score, and u_i represents prediction uncertainty.

$$\alpha + \beta + \chi + \lambda \quad (19)$$

This task enables higher clinical severity, higher risk, shorter deadlines to receive higher priority, during tasks related to high uncertainty, to minimize dangerous decision-making.

The monitored patients' physiological conditions are shown by HSI, which is evaluated from multiple clinical indicators derived during the feature extraction. This work consists of four types of physiological parameters, such as heart rate, corrected QT interval, oxygen saturation, and blood pressure.

$$HSI_i = \omega_1 F_{HR,i} + \omega_2 F_{QTc,i} + \omega_3 F_{SpO_2,i} + \omega_4 F_{BP,i} \quad (20)$$

From this equation, HRI, i is denoted as normalized heart rate deviation, QTc, i is indicated as corrected interval abnormality, SpO_2, i is the oxygen saturation deviation, and BP, i is represented as blood pressure abnormality.

The weighting coefficients ω_j can be computed as:

$$\sum_{j=1}^4 \omega_j = 1 \quad (21)$$

The weighting factors can be modified in accordance with clinical guidelines so that greater importance is given to parameters that have a strong association with critical cardiac events. To maintain clinical significance, the proposed TP-DSDRL model also integrates widely accepted physiological reference limits acquired from standard medical guidelines, such as the latest recommendations from European Resuscitation Council (ERC) [27] and American Heart Association (AHA) [28]. Based on these guidelines, the typical resting heart rate for adults lies between 60 and 100 beats per minute (bpm). A heart rate below 60 bpm is generally classified as bradycardia, whereas values exceeding 120 bpm may suggest severe tachycardia or increased cardiac strain. Accordingly, when the heart rate satisfies the corresponding task, the offering task offers higher severity in the HSI computation.

$$HR < 60 \text{ or } HR > 120 \quad (22)$$

In the same way, irregular ECG characteristics, including abnormal rhythms or extended QT intervals, may signal possible cardiac disorders. The QT prolongations that are clinically significant are specified as $QTc \geq 450 \text{ ms}$ in males and $QTc \geq 470 \text{ ms}$ in females, as stated in the latest cardiology literature [29]. As a result, the monitoring task related to these conditions is assigned a higher severity level. By incorporating clinically recognized threshold values, the prioritization strategy remains consistent with practical healthcare monitoring standards.

Since real-time healthcare systems must react quickly to sudden physiological variations, urgency also becomes an important factor. To capture this aspect, the deadline parameter is modeled using an inverse deadline sensitivity approach.

$$DL_i^{-1} = \frac{1}{\tau_i^{\text{deadline}} - \tau_i^{\text{arrival}}} \quad (23)$$

In this equation, τ_i^{arrival} is denoted as the task arrival time, and τ_i^{deadline} indicates the maximum delay for processing.

Even when the overall severity appears to be moderate, some physiological irregularities can still reflect a higher level of clinical risk. To account for these situations, an additional risk score is introduced in the model to represent such potentially critical situations.

$$R_i = \sum_{k=1}^m P_k \Phi_k \quad (24)$$

In this equation, Φ_k is the binary indicator that denotes the abnormal clinical condition, P_k denotes the significant weight related to the clinical condition, and m indicates the total number of monitored risk factors.

Typical indicators of elevated risk include the presence of arrhythmia, an extended QT interval, or irregular oxygen saturation levels. Predictions generated by machine learning models from physiological signals may sometimes involve uncertainty because of factors such as sensor noise or missing data. To maintain dependable task prioritization, the proposed framework incorporates a penalty component that accounts for uncertainty. In this approach, the level of uncertainty for each task is measured using an entropy-based confidence estimation method.

$$u_i = - \sum_{c=1}^C \rho_c \log(\rho_c) \quad (25)$$

From this equation,

$$\rho(\tau_i) = \begin{cases} \rho_1, & \rho_i \geq \tau_1 \\ \rho_2, & \rho_2 \leq \rho_i < \tau_1 \\ \rho_3, & \tau_3 \leq \rho_i < \tau_2 \\ \rho_4, & \rho_i < \tau_3 \end{cases} \quad (26)$$

Tasks identified with emergency priority (P1) are directly transmitted to the nearest edge server or cloud node to enable immediate processing. In contrast, tasks with lower priority levels are scheduled according to the availability of computational resources within the system.

4.6. State space construction

In reinforcement learning-based task offloading frameworks, the state space describes the current status of the network, available computing resources, and the characteristics of healthcare-related tasks as perceived by the learning agent. An appropriately structured state representation allows the model to interpret environmental conditions and determine efficient offloading strategies even when the system operates under dynamic circumstances.

Within the UAP-DSDRL model, the state space is formulated to incorporate both infrastructure-level resource information and medical task attributes. By combining these aspects, the agent can evaluate clinical urgency while simultaneously considering computational limitations when selecting suitable offloading actions.

At every decision time step, the overall system condition is represented through a multidimensional state vector that captures the relevant features required for the decision-making process.

$$\delta_t = \{\mathcal{R}_t^E, \mathcal{R}_t^C, HSI_t, \rho_t, LT_t, B_t\} \quad (27)$$

$\mathcal{R}_t^E$ Implies the available computational resources in the edge server, $\mathcal{R}_t^C$ signifies the available resources in the cloud server, HSI_t denotes the Health Severity Index, ρ_t indicates the task priority score, LT_t refers to the network latency, and B_t denotes the communication bandwidth.

By incorporating both the HSI and the priority score, the framework ensures that clinically critical healthcare tasks receive appropriate attention during the decision-making process. Meanwhile, system-related factors such as available computing resources, network latency, and bandwidth allow the agent to adjust its decisions according to changing communication and computation conditions. Through this combined representation, the model gains a holistic understanding of the cloud-edge healthcare environment. Consequently, the UAP-DSDRL agent can learn effective task offloading strategies that reduce processing delay while guaranteeing the timely execution of high-priority medical tasks.

4.7. Task offloading using UAP-DSDRL

Once task priorities are determined and the system state space is

defined, the proposed framework proceeds with intelligent task offloading through the UAP-DSDRL model. The primary goal of this component is to identify the most suitable execution location for each healthcare task. By doing so, the framework aims to reduce processing latency and energy usage while ensuring that medically critical tasks are allocated immediate computational support. The architecture of the DSDRL Technique for Healthcare task offloading is illustrated in Fig. 4.

The task offloading problem is designed as a Markov Decision Process (MDP), which is denoted as a tuple (SS, AS, RW, π) . In this, SS indicates the state space, AS denotes the action space, RW represents the reward function, and π signifies the decision policy. Every decision step t , the agent determines the current system state SS_t and chooses an action AS_t that evaluates the execution of the task.

The action space can be expressed as:

$$AS_t \in \{\alpha_1, \alpha_2, \alpha_3\} \quad (28)$$

In this equation, α_1 denotes the local execution at the device, α_2 indicates task offloading at the edge server, and α_3 represents the task offloading to the cloud server. Once an action is performed, the environment moves to the next state SS_{t+1} , and the agent receives a reward that evaluates the quality of the chosen decision.

The optimal action-value function is estimated using Q-learning that formulated as:

$$Q(SS_t, AS_t) = R_t + \beta \max_A Q(SS_{t+1}, AS) \quad (29)$$

From this equation, $Q(SS_t, AS_t)$ refers cumulative reward for taking action AS_t in state SS_t , R_t represents the immediate reward, and β denotes the discount factor.

Since the system operates in a high-dimensional state environment, directly estimating Q-values becomes challenging. To address this, the proposed TP-DSDRL model utilizes an SCNN as a function predictor. This network learns compact and meaningful representations of the system state by identifying relationships among network conditions, available resources, and healthcare task priorities.

The SCNN model is composed of three convolutional layers followed by pooling and fully connected layers. The first convolutional layer contains 32 filters with a kernel size of 3×3 and employs the ReLU activation function to extract fundamental features from the input state. After this, a max-pooling operation with a pooling size of 2×2 window is used to decrease the feature dimensionality and enhance generalization capability. The second convolutional layer uses 64 filters with the same kernel size 3×3 to capture more complex feature interactions. Finally, a third convolutional layer with 128 filters of size 3×3 is used to learn deeper representations of the system state. Fig. 5 displays the structure of an SCNN.

To improve training stability and limit the risk of overfitting, a global average pooling layer is applied to summarize the extracted feature maps before they are forwarded to a fully connected layer with 256 neurons. The final output layer then generates Q-values for each possible task offloading option, allowing the agent to choose the execution strategy that yields the highest expected benefit.

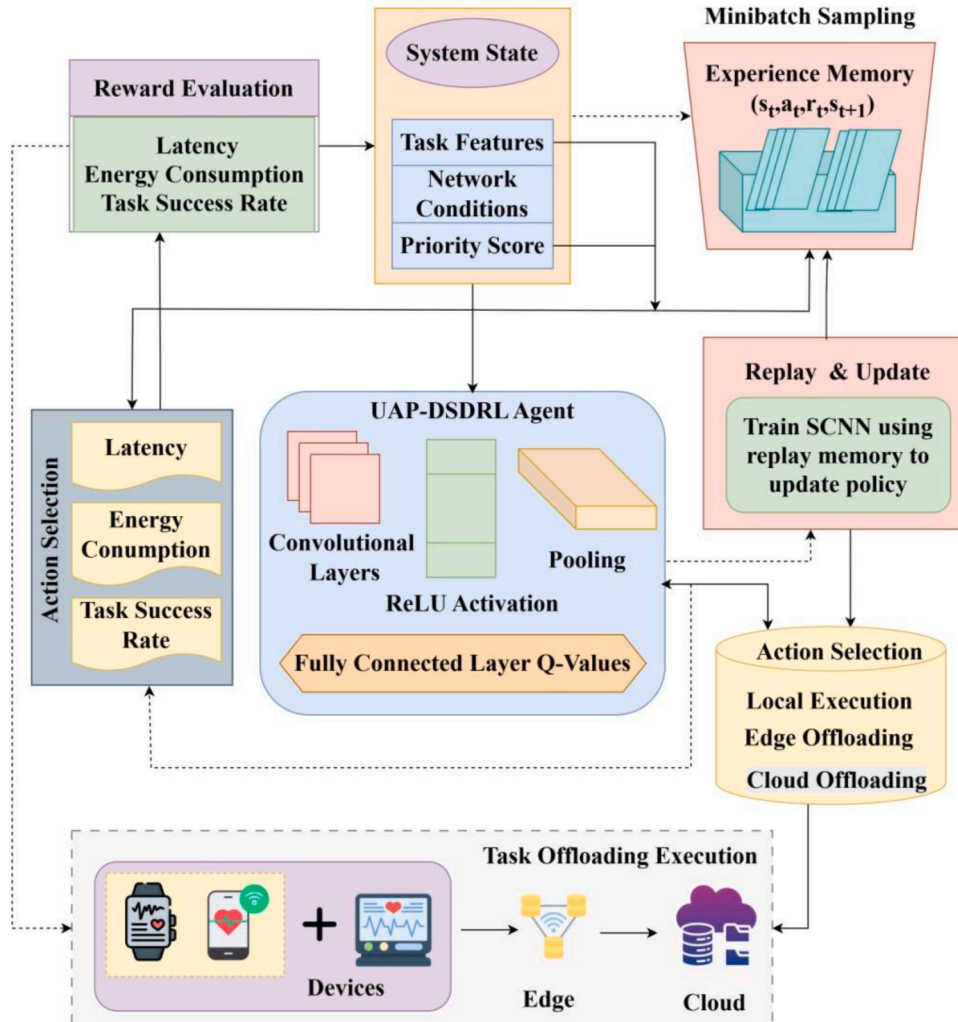


Fig. 4. Architecture of the DSDRL method for Healthcare task offloading.

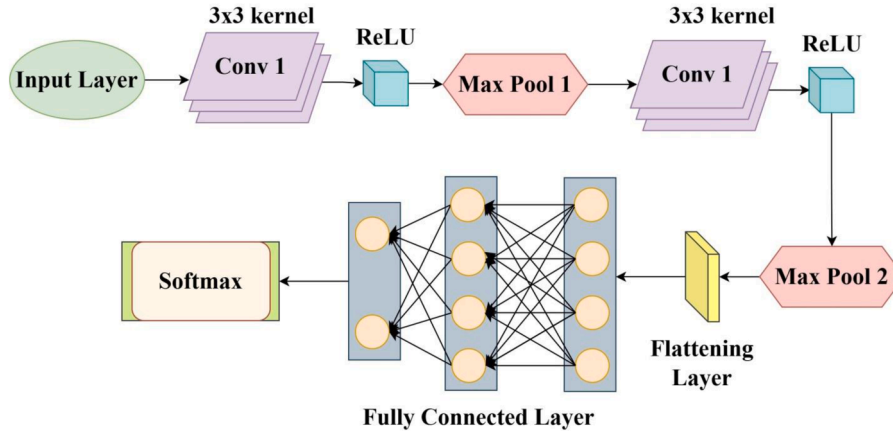


Fig. 5. Structure of SCNN.

To make the decision process more reliable, the framework integrates an uncertainty-aware component within the policy network. The uncertainty associated with the predicted Q-values is measured through variance-based analysis, which helps the agent avoid potentially risky offloading choices when the prediction confidence is low. Additionally, an adaptive ϵ -greedy exploration strategy is used during training to maintain a balance between exploration and exploitation. In this strategy, the agent selects the best-known action with probability $1 - \epsilon$, while a random action is chosen with probability ϵ .

By continuously interacting with the cloud-edge healthcare environment, the UAP-DSDRL agent gradually learns an adaptive task offloading policy capable of responding to variations in network conditions, available computational resources, and patient severity levels. Consequently, the proposed method promotes efficient utilization of system resources while ensuring that high-priority healthcare monitoring tasks are processed without delay.

4.8. Multi-objective reward evaluation

In the proposed TP-DSDRL model, the reinforcement learning agent identifies optimal task loading decisions by maximizing a carefully formulated multi-objective reward function. Healthcare task scheduling typically involves several competing goals, including reducing latency, minimizing energy usage, lowering operational expenses, and guaranteeing priority-aware service for critical tasks. To address these objectives simultaneously, the reward formulation combines these factors into a single optimization criterion.

At every decision step d , the reward value RW_d measures the effectiveness of the chosen action by evaluating aspects such as task completion performance, availability of computational resources, and overall system workload distribution. The reward function is expressed as follows:

$$RW_d = \alpha \left(\frac{\eta_{d+1}^E}{\eta_d^E} \right) + \beta \left(\frac{\eta_{d+1}^A}{\eta_{\max}^A} \right) + \chi \left(\frac{\nu_d^{ave}}{\nu_{\max}^{ave}} \right) \quad (30)$$

In this equation, α , β , and χ indicate the weighting parameters satisfying.

$$\alpha + \beta + \chi = 1 \quad (31)$$

These weighting coefficients regulate how much influence each objective factor has during the learning process.

The component η_{d+1}^E refers to the number of emergency healthcare tasks that are successfully completed at the following decision step $(d + 1)$. These tasks correspond to high-priority monitoring operations identified through the Health Severity Index (HSI). A larger value of η_{d+1}^E represented that the system is effectively giving precedence to patients with critical medical conditions.

Similarly, the parameter η_{d+1}^A indicates the number of computing nodes that remain free after tasks have been assigned at the step $(d + 1)$. This value represents the residual computational capacity within the edge-cloud infrastructure. Preserving a higher level of node availability helps to prevent system congestion and allows the platform to accommodate additional incoming tasks efficiently.

The variable ν_d^{ave} represents the mean processing speed of the computing nodes that are active during the decision step d , indicating the system's overall computational performance. It is calculated as the average processing capability of all active nodes:

$$\nu_d^{ave} = \frac{1}{K} \sum_{k=1}^K \nu_k \quad (32)$$

In this equation, K denotes the total number of active computing nodes and ν_k corresponds to the processing speed of the k^{th} node.

By integrating these elements, the reward formulation directs the TP-DSDRL agent toward learning a policy that gives preference to urgent healthcare tasks, maintains balanced use of available resources, and enhances overall task processing efficiency. This multi-objective reward structure supports effective and dependable task offloading within dynamic cloud-edge healthcare management systems.

4.9. Uncertainty-aware policy update

The inaccurate task offloading decisions cause an increase in the latency rate and incorrect permutation of computational resources due to changes in the healthcare environment. The proposed TP-DSDRL model integrates an uncertainty-aware policy update mechanism among reinforcement learning process to help increase the decision reliability. The integrated mechanism enables the agent to calculate the forecasted action strength and reduce the high-risk offloading decision while uncertainty is high.

The uncertainty in action-value estimation is determined using the variance of Q-values generated by the Stacked SCNN estimator. Let $Q_k(S_d, A_d)$ represent the Q-value predicted by the k^{th} estimation instance for state S_d and action A_d . The average Q-value is written as:

$$\bar{Q}(SS_d, AS_d) = \frac{1}{K} \sum_{k=1}^K Q_k(SS_d, AS_d) \quad (33)$$

From this equation, K signifies the total number of Q-value estimators.

Based on this average value, the uncertainty corresponding to a specific action is determined using a variance-based formulation:

$$US_d = \frac{1}{K} \sum_{k=1}^K (Q_k(S_d, A_d) - \bar{Q}(S_d, A_d))^2 \quad (34)$$

This equation represents that US_d indicates the uncertainty score in the decision step US_d . A larger value of US_d denotes the lower confidence in the predicted action.

The integration of uncertainty into policy learning, the upgrade of the Q-value can be formulated as:

$$Q_n(SS_d, AS_d) = Q_o(SS_d, AS_d) + \chi \left[R_d + \mu \max_{A'} Q(SS_d + 1, A') - Q_o(SS_d, AS_d) \right] (1 - U_d) \quad (35)$$

In this equation, χ indicates the learning rate, μ denotes the discount factor, R_d signifies the reward, and AS_d implies the action.

The factor $(1 - US_d)$ decreases the influence of the update whenever the uncertainty level is high, thus limiting unstable adjustments in the policy. Through this mechanism, the reinforcement learning agent is encouraged to favor actions with higher reliability, which helps maintain safe and consistent task scheduling decisions within cloud-edge healthcare environments.

Furthermore, an adaptive ϵ -greedy exploration strategy is used to maintain a balance between exploration and exploitation during training. In this approach, the exploration probability gradually decreases over time according to:

$$\epsilon_d = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min}) e^{-\lambda d} \quad (36)$$

From this equation, $\epsilon_{\max}$, and $\epsilon_{\min}$ denote the maximum and minimum exploration rates, respectively, and λ determines the decay speed.

By combining uncertainty estimation with this adaptive exploration mechanism, the TP-DSDRL model achieves more stable policy learning and enhances the reliability of task offloading decisions even under changing network conditions and fluctuating workloads.

4.10. Closed-loop feedback mechanism

To support dependable decision-making in dynamic healthcare environments, the proposed TP-DSDRL framework incorporates a closed-loop feedback mechanism that continuously refines the learning model using real-time system observations and stakeholder input. In real-world healthcare settings, clinicians, monitoring personnel, and system administrators may modify task priorities depending on patient status, emergency notifications, or network congestion levels. Incorporating such real-time feedback allows the reinforcement learning model to adjust its policy in response to changing operational demands.

Within the Deep Q-Network (DQN) learning framework, the model parameters are optimized by minimizing a loss function that captures the difference between the predicted Q-value and the corresponding target value. At decision step x , the loss is defined as:

$$LS(\theta_d) = (Q(S_d, A_d; \theta_d) - b_d)^2 \quad (37)$$

In this equation, $Q(S_d, A_d; \theta_d)$ indicate the predicted Q-value produced by the SCNN-based policy network with parameters x , while x represents the target Q-value used to guide the training process.

The target Q-value is determined according to the Bellman update formulation:

$$b_d = RW_d + \mu \max_{A'} Q(SS_d + 1, A'; \theta_d^-) \quad (38)$$

From this equation, RW_d is the reward function received after acting, μ indicates the discount factor, and $Q(SS_d + 1, A'; \theta_d^-)$ indicates the Q-value predicted by the target network for the next state $SS_d + 1$.

The proposed closed-loop feedback mechanism incorporates real-time input from stakeholders by dynamically modifying the reward signal and task priority parameters based on updated system observations. For example, clinicians may flag certain patient conditions that demand urgent processing, whereas system administrators may report issues such as network congestion or limited resource availability. These

inputs are integrated into the learning process by adjusting the reward value RW_d and the priority weights assigned to healthcare tasks.

In addition, important system indicators—including latency, node utilization, and task completion rates—are continuously monitored and incorporated into the state representation. This feedback information enables the reinforcement learning agent to progressively refine its policy during training by adjusting to changing operational conditions within the system.

During this closed-loop learning process, the proposed TP-DSDRL model increases the decision policy, enabling a task offloading method still responsive to the patients' requirements, availability of system resources, and limitation of operations, all of which are improved consistently. Overall, the proposed TP-DSDRL attained adaptive, reliable, and scheduling medical aware task in cloud-edge healthcare environments.

5. Experimental evaluation results

This section describes the experimental validation results, including the simulation setup, parameter settings, and evaluation indicators. Then the analysis of the results evaluates the effectiveness of the TP-DSDRL approach.

5.1. Experimental setup

System Configuration: This research utilizes a robust hardware configuration including an Intel® Core™ i7-6700HQ CPU, 16GB of Random Access Memory (RAM), and an NVIDIA GeForce GTX 960 M Graphics Processing Unit (GPU), all operating on Ubuntu 16.04.6 LTS (64-bit). This research uses CloudSim software, which is designed for modeling and simulating cloud infrastructures.

5.2. Dataset description

• Healthcare dataset

The Healthcare dataset (<https://www.kaggle.com/datasets/prasad22/healthcare-dataset>) contains 10,000 patient records, whereas these records indicate synthetic patient healthcare records individually. This record is categorized into different types includes patient demographics, medical conditions, and admission details. This dataset it is designed for use in the field of education and commercial use. It carries artificial data only, and does not carry real patient data.

• Healthcare_IOT_Data

The Healthcare IoT dataset provides a comprehensive collection of data collected from various IoT devices used in healthcare settings. This dataset typically includes information on patient vitals, device statuses, and other health metrics, which are aimed at enabling research and development in health monitoring, predictive analytics, and personalized treatment options. Generally, it encompasses features such as timestamps, device IDs, sensor readings (like heart rate, temperature, blood pressure), and patient demographics. Such datasets are essential for developing algorithms to enhance patient care, predictive models for disease progression, and improving overall health outcomes through data-driven decision-making in clinical environments. This dataset is available at <https://www.kaggle.com/datasets/ziya07/healthcare-iot-data>.

5.3. Hyperparameter configuration

Table 1 shows the hyperparameter configuration for the proposed TP-DSDRL model. To achieve reliable training and better performance of the proposed TP-DSDRL framework, the hyperparameters were determined through detailed experiments and ablation analysis. A grid search

Table 1
Hyperparameter configuration of TP-DSDRL method.

Hyperparameter	Optimal Value
Number of IoT Devices	150
Edge Servers	5
Cloud Servers	1
Training Episodes	2000
Time Step	1 s
Learning Rate	0.0003
Discount Factor (γ)	0.99
Batch Size	64
Replay Buffer Size	100,000
Target Network Update Frequency	300 steps
Optimizer	Adam
Initial Exploration (ϵ)	1.0
Final Exploration (ϵ)	0.01
ϵ Decay Rate	0.995
Q-Variance Threshold	0.03
Confidence Window Size	15
Uncertainty Penalty Coefficient	0.2
Convolution Layers	3
Filters per Layer	64
Kernel Size	3×3
Activation Function	ReLU
Dropout Rate	0.3
BiLSTM Hidden Units	128
BiLSTM Layers	2
Transformer Heads	4
Transformer Layers	1
Attention Dropout	0.2
Latency Weight (w1)	0.4
Energy Weight (w2)	0.3
Cost Weight (w3)	0.2
Reliability Weight (w4)	0.1
Emergency Threshold (HSI)	≥ 0.80
High Priority Range	0.60–0.79
Medium Priority Range	0.40–0.59
Low Priority Range	< 0.40

approach was adopted to identify suitable values for important parameters such as the learning rate and dropout rate. During tuning, the learning rate was evaluated with different values (0.0001, 0.0003, 0.0005, and 0.001). Among these, 0.0003 provided a balanced outcome by ensuring stable training while maintaining a reasonable convergence speed. When higher learning rates were used, the reward values fluctuated significantly, whereas smaller values resulted in slower convergence of the model. In the same way, different dropout configurations (0.0, 0.3, and 0.5) were tested to control overfitting. A dropout rate of 0.3 produced the most effective results, as it helped reduce overfitting while still preserving the model's ability to generalize. The remaining hyperparameters, such as the discount factor, batch size, replay buffer capacity, and the update interval of the target network, were selected based on commonly adopted DRL practices and further verified through preliminary trials. With these final settings, the framework achieved stable convergence along with improved task offloading efficiency, lower latency, and better energy utilization.

5.4. Evaluation indicators

The evaluation indicators employed for analyzing the TP-DSDRL method performance are presented in this section. The indicators comprise Latency, execution time, energy consumption, processing time, throughput, average resource utilization, average task offloading delay, and energy efficiency.

Latency (L_{cy}): Latency refers to the time delay between request initiation and the corresponding response or action. In task offloading, it includes the time taken to transfer data to the cloud, process it, and return the results to the edge device. The Latency is computed using the following equation.

$$LT = \frac{1}{N} \sum_{i=1}^N (\tau_i^{fin} - \tau_i^{str}) \quad (39)$$

From the above equation, the average latency is signified as LT , the total number of tasks is indicated as N , the task generation time is denoted by τ_i^{str} , and the task completion time is represented by τ_i^{fin} .

Execution time (ET): This refers to the total time taken to complete a task once it has been offloaded to the cloud or edge server.

$$ET = \frac{C_i}{f_j} \quad (40)$$

where C_i stands for the computational workload of the i_{th} task, and f_j represents the processing speed of the j_{th} node.

Energy consumption (E_c): It calculates the sum of energy consumed by the IoT devices for task of computation and transmission.

$$EC = \frac{1}{N} (EC_i^{cmp} + EC_i^{trs}) \quad (41)$$

In the above equation, the total energy consumption is signified by EC , the energy employed for computation is denoted as EC_i^{cmp} , the energy utilized for the transmission task is represented as EC_i^{trs} , and moreover, the efficiency of the device battery is enhanced by the reduced energy consumption.

System Throughput: The number of healthcare tasks that are executed in a specific interval of time is measured using the system throughput. The formulation for the throughput is as follows:

$$ThP = \frac{N_{proc}}{T_p} \quad (42)$$

From Eq. (42), the system throughput is indicated as ThP , the number of processed tasks is signified as N_{proc} , and the entire processing time is denoted as T_p .

Processing time (PT): Processing time is the duration required to process a healthcare task exclusively on a node, without considering communication delays.

$$PT = \frac{D_i}{R_j} \quad (43)$$

where D_i stands for the data size of the i_{th} task, and R_j represents the data processing rate of the j_{th} node.

Average resource utilization (R_{avg}): This metric reflects how effectively the cloud and edge resources are being used during task execution. It takes into account CPU, memory, and bandwidth usage. High resource utilization signifies efficient use of available resources.

$$R_{avg} = \frac{1}{N} \sum_{j=1}^N \frac{R_j^{used}}{R_j^{total}} \times 100 \quad (44)$$

where R_j^{used} indicates resource used by node j , N stands for total number of nodes, and R_j^{total} represents the total resources of the node j .

Average task offloading delay ($T_{offload}$): This refers to the delay experienced when a task is offloaded from a local device to the cloud or edge server before execution. Factors influencing this delay include network latency and the time taken for data transfer.

$$T_{offload} = \frac{S_i}{B_j} + T_{queue} \quad (45)$$

where S_i represents the size of the offloaded task, bandwidth of the communication link is denoted by B_j , and T_{queue} stands for average queuing delay at the offloading node.

Energy efficiency (EF): Energy Efficiency indicates the ratio of successful task completions to the total energy consumed. Higher values imply better system performance.

$$EF = \frac{N_{completed}}{E_{total}} \quad (46)$$

where $N_{completed}$ stands for the number of completed healthcare tasks, and E_{total} represents the total energy consumption during the period.

5.5. Performance analysis of the TP-DSDRL method

The performance analysis of the proposed TP-DSDRL method in terms of several metrics using the two different datasets for intelligent task offloading in the management of healthcare in cloud edge environments is presented in this section.

System Performance under increasing IoT devices against Latency, Energy Consumption, and Task Success Rate is shown in Fig. 6. The increase in the number of IoT devices from 50 to 150 progressively maximizes the average latency because of greater processing loads and data transmission in the cloud-edge environments, but in the proposed method, the latency is relatively low, describing the efficiency in handling the developing healthcare workloads. The usage of energy increases from 11.2 J to 24.6 J when there is a rise in IoT devices, based on the increased communication and computational demands. The task success rate is achieved above 95 % when analyzing the number of task thereby signifying strong reliability of the system.

Fig. 7 illustrates the Task Success Rate analysis of the proposed TP-DSDRL over Network Bandwidth. An improved task success rate is achieved, ranging from 94 % to 98.5 %, when the network bandwidth ranges from 5 Mbps to 25 Mbps, signifying that an increase in bandwidth allows faster transmission of data and minimizes the congestion of the network while offloading the healthcare task.

The Task priority level analysis of the proposed TP-DSDRL over Response time is displayed in Fig. 8. A minimum response time of 70 ms is attained by the emergency task, assuring faster execution of the critical data of the patient. The response time is regularly maximized where the priority level is reduced from high to low, signifying urgent tasks are prioritized efficiently for faster processing.

Fig. 9 shows the Task utilization estimation of the proposed TP-DSDRL against the number of tasks. The utilization of resources rises to 84 % when the number of tasks is 500, signifying effective computing resource allocation in the cloud edge environment. The enhancement is based on the efficient distribution of workload and adaptive task offloading.

The Impact of the uncertainty-aware module of the proposed TP-DSDRL is illustrated in Fig. 10. The rise of system load from 20 % to 100 % maximized the offloading decision error rate. Lower error rates are attained consistently using the uncertainty-aware module of the proposed UAP-DSDRL method when compared to handling the model in the absence of uncertainty. This describes that the accuracy and reliability of the task offloading decisions are enhanced over the dynamic condition of the system while integrating the uncertainty awareness.

The Learning Performance of TP-DSDRL in terms of Average Reward against Training Episodes and Training Loss against Training Episodes is

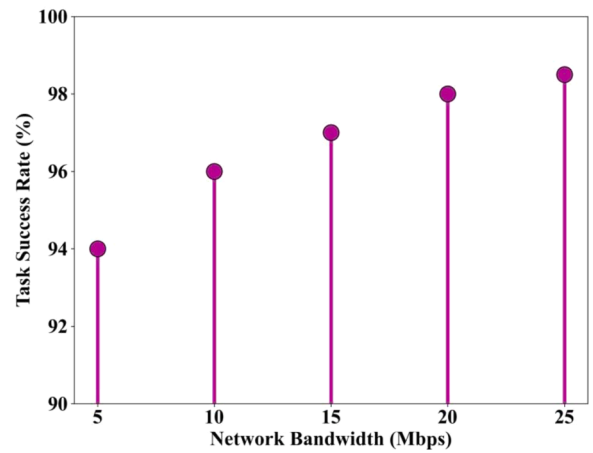


Fig. 7. Task Success Rate analysis of the proposed TP-DSDRL over Network Bandwidth.

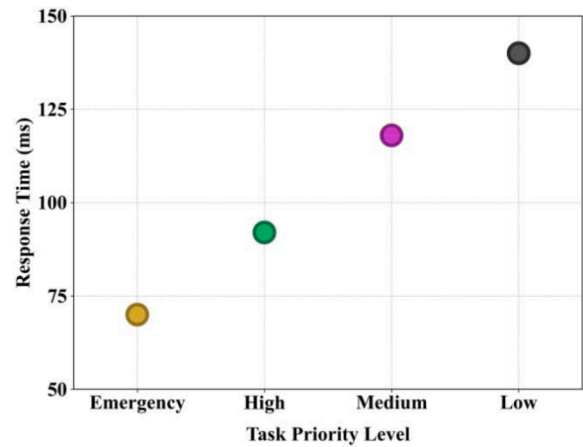


Fig. 8. Task priority level analysis of the proposed TP-DSDRL over Response time.

displayed in Fig. 11. The rise in the number of training episodes from 200 to 2000 steadily maximizes the average reward, signifying that the proposed method learns the offloading strategy progressively using the continuous correlations with the environments. The reward improvement describes better resource utilization and decision making, resulting in stable convergence learning over time. In training loss, it minimizes regularly due to the raise of episodes. From 0.81 to 0.24, the loss range reduces, representing an enhanced accuracy of the model in reducing the prediction error efficiently.

The Healthcare Priority-Based Performance of the proposed TP-DSDRL method based on (a) Response Time and Task Priority and (b)

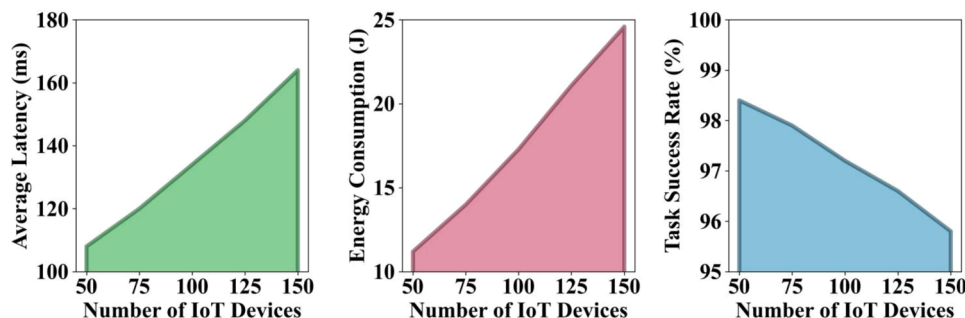


Fig. 6. System Performance under increasing IoT devices against Latency, Energy Consumption, and Task Success Rate.

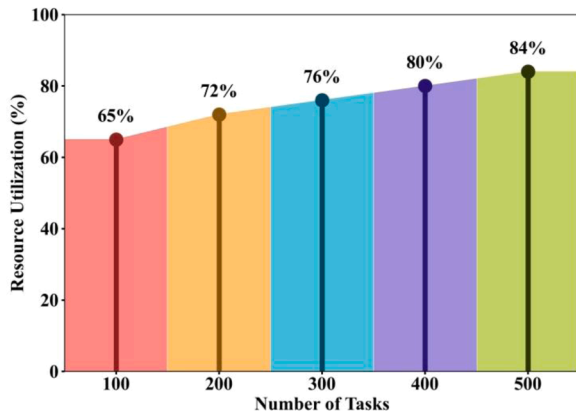


Fig. 9. Task utilization estimation of the proposed TP-DSDRL against the number of tasks.

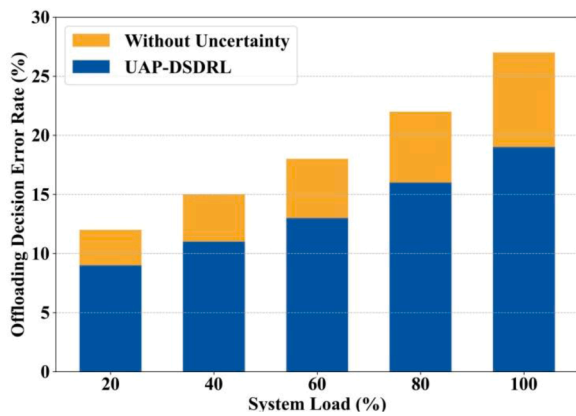


Fig. 10. Impact of the uncertainty-aware module of the proposed TP-DSDRL method.

Task offloading Distribution and Task Priority is shown in Fig. 12. Based on the healthcare priority level task, the response time changes, and the emergency tasks attains minimum of 68 s response time, ensuring faster handling of the patient's data. Meanwhile, the tasks from high to low generates gradual increase in the response time. These results ensure that the emergency healthcare tasks are prioritized efficiently by the

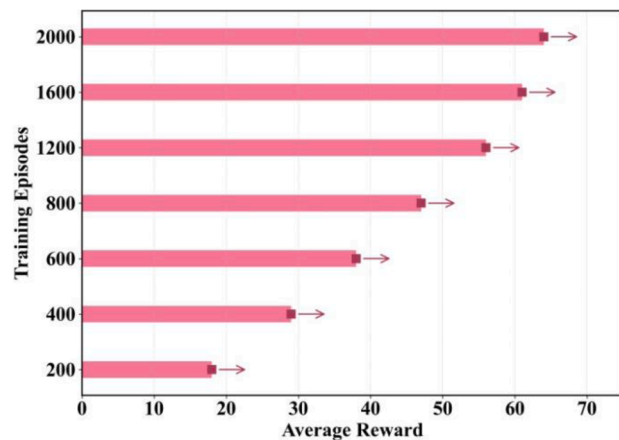
proposed method for quick processing. In this, the task offloading distribution also varies, and at the edge layer, the high-priority and emergency tasks are processed for assuring quick response time. The cloud and device resources handle the low and medium priority tasks regularly for balancing the load of computation during the maintenance of timely healthcare task processing.

The task offloading scalability system for the proposed TP-DSDRL method, in comparison of the number of tasks with (a) Throughput and (b) Edge Resource Utilization, is displayed in Fig. 13. When the number of task increase from 100 to 600, the throughput maximizes, signifying that more tasks per second are executed along with a greater workload. The proposed method's capabilities are described in which the greater volumes of tasks are handled efficiently. The rise of number of task increase the utilization of edge resources between the range of 100 to 600. From 64 % to 88 %, the utilization has increased, reflecting the effective available edge resources utilization, which happens based on the efficient balancing of workload and adaptive task offloading.

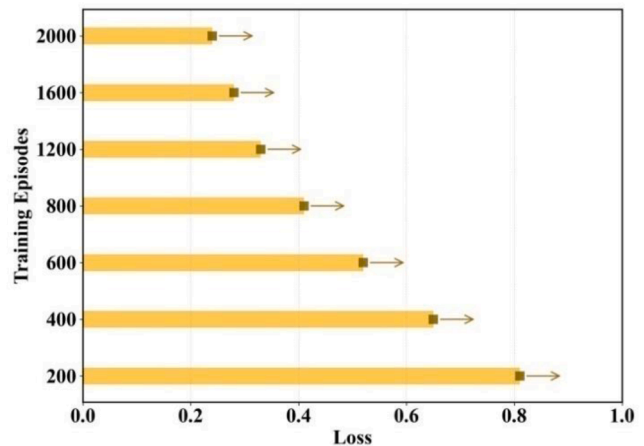
Task Processing Histogram of the proposed TP-DSDRL method in terms of latency and Energy Consumption is displayed in Fig. 14. In the 110–170 ms latency range, most of the healthcare tasks are accomplished, signifying faster processing and effective scheduling of tasks in the cloud edge environment. Beyond 200 ms, fewer tasks undergo greater latency when there is an increase in the system load. Entirely, the proposed method sustains low and stable processing delay, which is necessary for monitoring the real-time healthcare application. Meanwhile, within the energy range of 11–17 J, most of the tasks utilizes indicating that the IoT device's computational burden is reduced efficiently using the proposed offloading method. Higher energy is consumed by the minimum tasks during the requirement of additional computation or communication, signifying that the processing of energy efficient task are maintained successfully for the IoT healthcare and wearable devices, prolonging the life of the battery.

Fig. 15 illustrates the histogram of HSI over the patient risk level distribution. A moderate severity of 0.4–0.6 range is produced by most of the cases, whereas a high risk at 0.8–0.10 range is generated because of a smaller number of cases. Based on this distribution, the patient health conditions are captured efficiently using the proposed model, and the dynamic task prioritization is allowed, in which the healthcare tasks are important for receiving instant attention.

The histogram of Training Reward Distribution of the proposed TP-DSDRL method is shown in Fig. 16. Between 40 and 60, the reward values are included, signifying that effective offloading policies are identified consistently by the learning agent. The model attains greater reward values, providing efficient decision-making and stable



(a)



(b)

Fig. 11. Learning Performance of TP-DSDRL in terms of (a) Average Reward and Training Episodes and (b) Training Loss and Training Episodes.

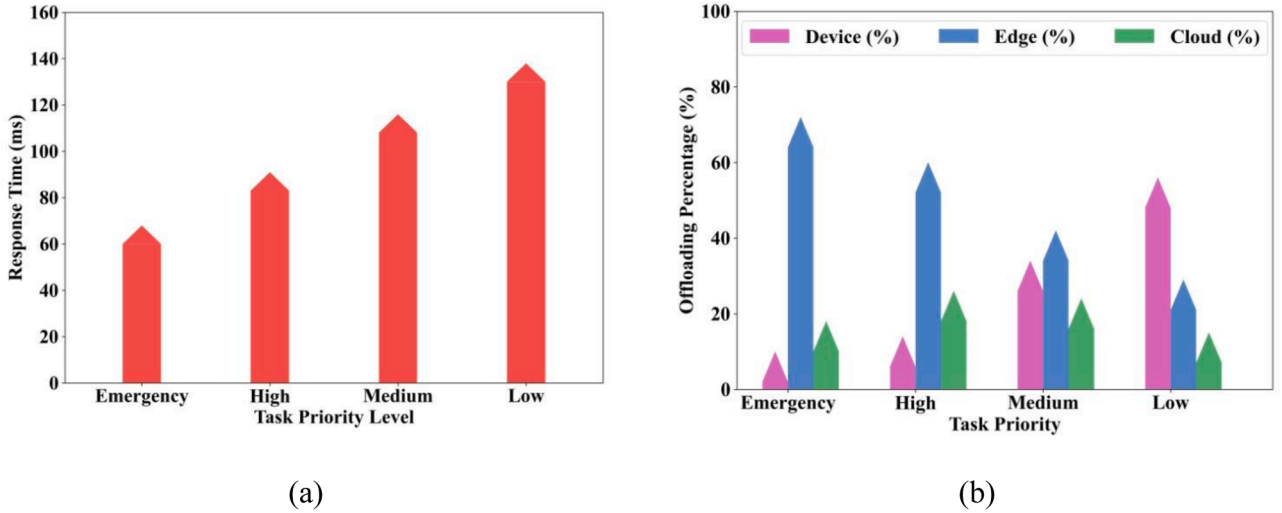


Fig. 12. Healthcare Priority-Based Performance of the proposed TP-DSDRL method based on (a) Response Time and Task Priority and (b) Task offloading Distribution and Task Priority.

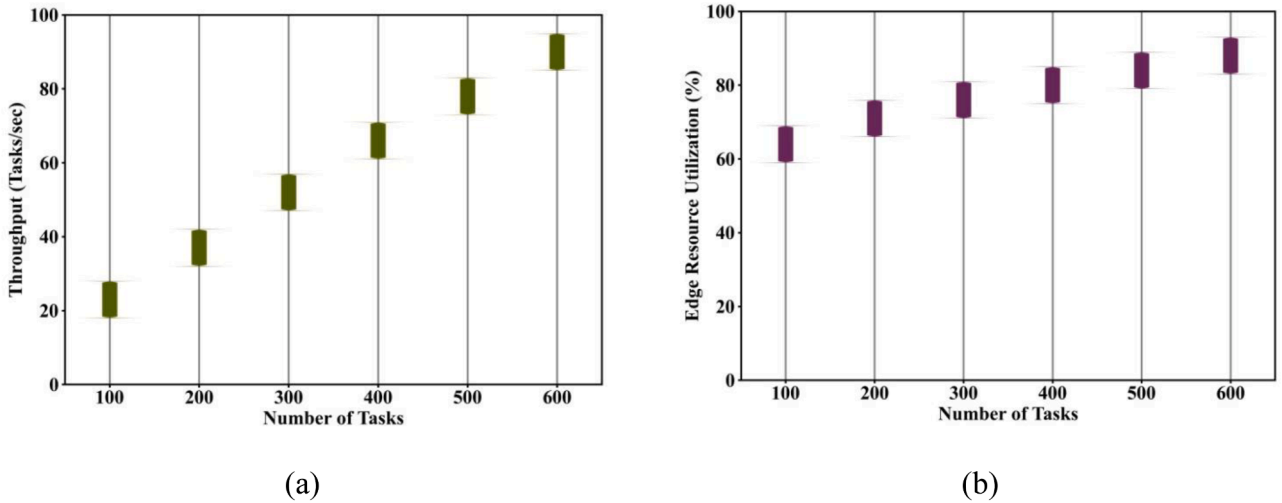


Fig. 13. Task offloading scalability system for the proposed TP-DSDRL method in comparison of Number of tasks with (a) Throughput and (b) Edge Resource Utilization.

convergence, resulting in minimized system latency and optimized utilization of resources in the healthcare environment based on edge and cloud systems.

Table 2 offers the performance of the TP-DSDRL technique over healthcare-related datasets. The findings clearly reveal the exceptional performance of the proposed method in task-offloading-based healthcare management over diverse performance parameters. Moreover, the Healthcare IoT dataset demonstrated better performance compared to the Healthcare dataset, which clearly reveals the improved predictive abilities. Furthermore, the average values of the various metrics help in obtaining the total effectiveness of TP-DSDRL in dealing with healthcare management through task offloading.

The probability density plot of the proposed TP-DSDRL method over the task processing latency distribution is illustrated in Fig. 17. Around 140 ms, the curve of the density tops, signifying that most tasks of healthcare are executed. Stable performance of the system along with the reduced extreme delays is demonstrated by the sleek bell-shaped distribution, verifying the efficiency of the proposed offloading strategy and task scheduling in the healthcare cloud and edge environment.

The training reward convergence of the proposed TP-DSDRL method under varied learning rates is displayed in Fig. 18. While training the

epoch early, the reward value significantly maximizes when the environments are explored, and the optimal task offloading strategies are optimized using the agent. Gradually, the reward stabilizes when the number of epochs maximizes, signifying that the optimal policy has converged by the reinforcement learning agent. The training stability and effectiveness of the proposed TP-DSDRL method are confirmed by the stable reward trend.

5.6. Comparative analysis

This section discusses the result analysis of the proposed approach compared to other existing methods: EA-MFO [11], DTH-ATB-MAPPO [12], DEEO [13], DPTARA [14], OO-DRL [15], DRL-LOA [17], and PBDLB [18]. The comparison highlights the improvements in various performance indicators achieved by the TP-DSDRL technique.

Fig. 19(a) and (b) illustrate a comparative analysis of the proposed TP-DSDRL technique against existing techniques for Latency and Execution Time. This analysis illustrates that the proposed TP-DSDRL method has the lowest latency of 170 ms. This result reveals the effectiveness of the TP-DSDRL method in providing efficiency in the processing time. Moreover, the proposed TP-DSDRL technique attains an

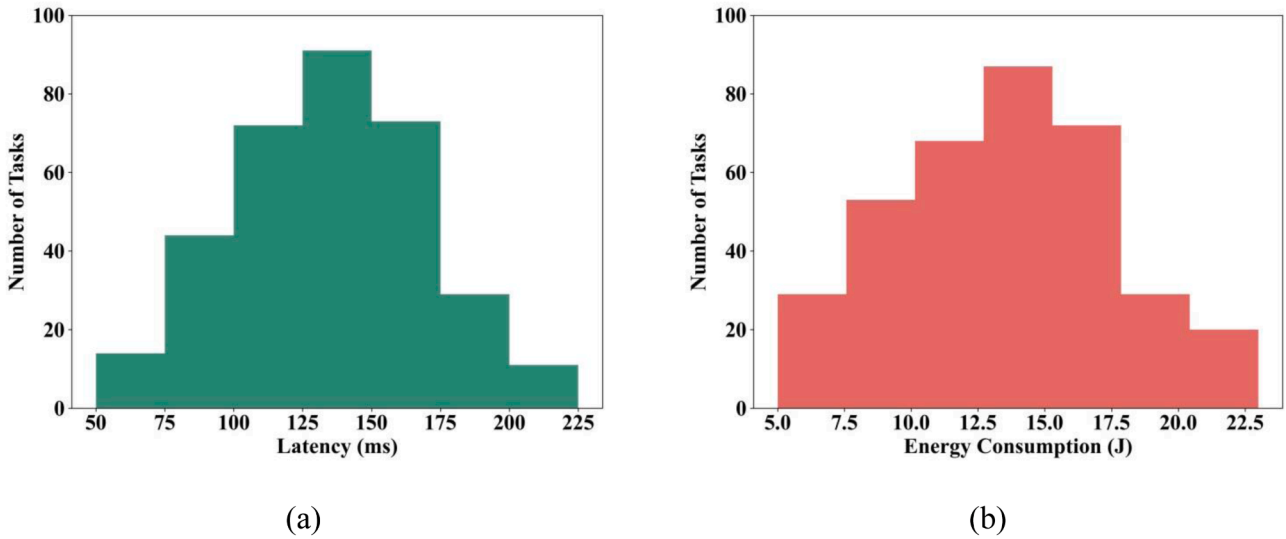


Fig. 14. Task Processing Histogram of the proposed TP-DSDRL method over (a) latency and (b) Energy Consumption.

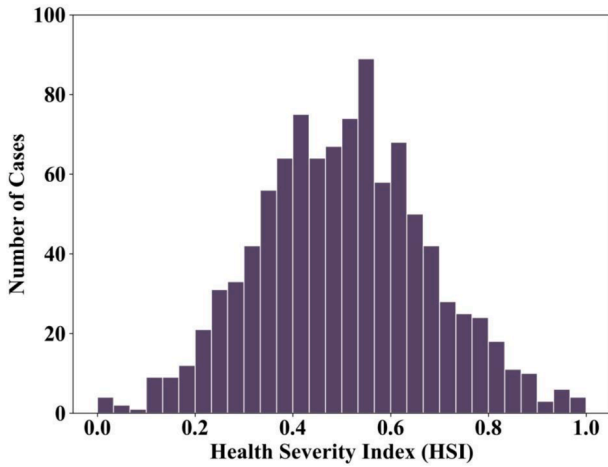


Fig. 15. Histogram of HSI over the patient risk level distribution.

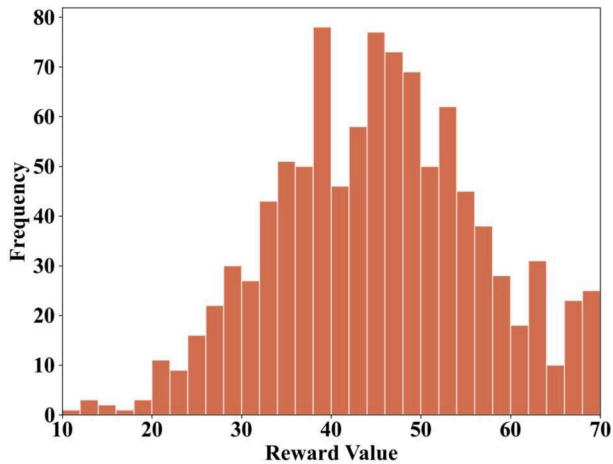


Fig. 16. Histogram of Training Reward Distribution of the proposed TP-DSDRL method.

execution time of 620 ms, which is lower compared to other methods, resulting in faster completion of the offloading task in the management

Table 2

The efficiency of two varieties of healthcare dataset analysis applying the TP-DSDRL technique.

Measures	Datasets		Average Values
	Healthcare dataset	Healthcare_IoT_Data	
Latency (ms)	190	150	170
Execution Time (ms)	650	590	620
Energy consumption (kJ)	530	450	490
System Throughput (Tasks/sec)	95	85	90
Processing Time (ms)	0.48	0.54	0.51
Average resource utilization (%)	92	76	84
Average task offloading delay (ms)	24	18	21
Energy efficiency (%)	97.35	98.71	98.03

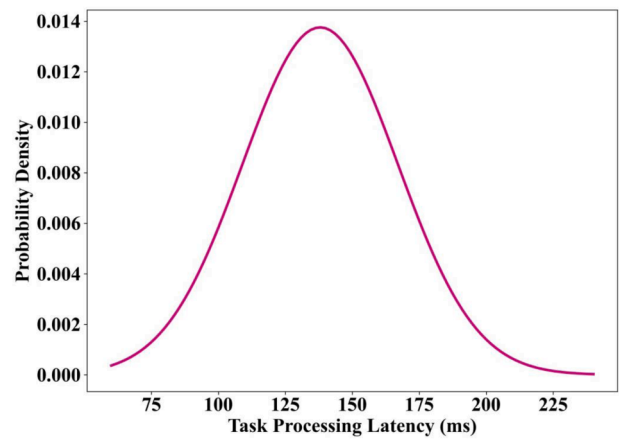


Fig. 17. Probability density plot of the proposed TP-DSDRL method over the task processing latency distribution.

of healthcare records.

Comparison of the proposed TP-DSDRL technique against existing techniques in terms of energy consumption and system throughput over different numbers of tasks is shown in Fig. 20. Over 250 tasks, the proposed TP-DSDRL consumes 490 kJ of energy, whereas the energy

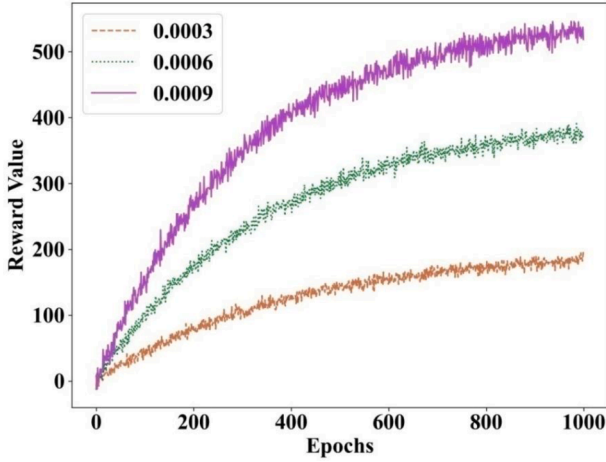
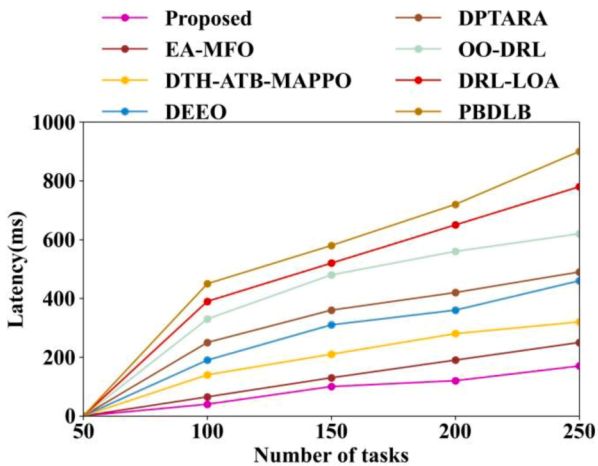


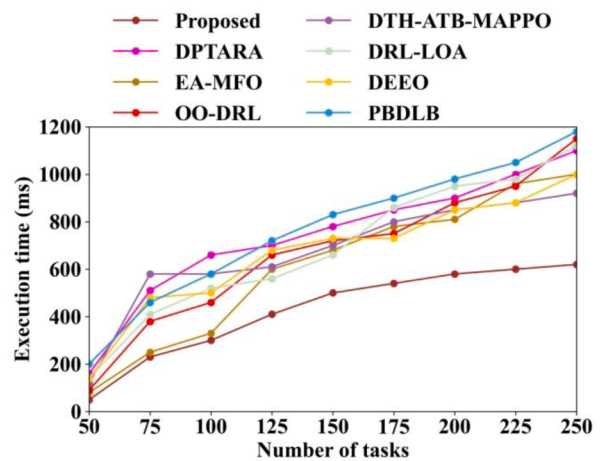
Fig. 18. Training reward convergence of the proposed TP-DSDRL method under varied learning rates.

consumption ratio for the other techniques is higher. When the energy consumption is optimized, it provides an increased life for devices, thereby providing customer satisfaction. Since the proposed TP-DSDRL consumes less energy significantly, it not only provides an improved life for the cloud-edge device but also makes the overall network environment sustainable. The system throughput of the proposed method attains 90 tasks/sec in 600 tasks, describing higher values when compared with the existing methods. The higher the throughput signifies efficient ability to handle a greater volume of tasks.

Comparison of the proposed TP-DSDRL technique against existing techniques in terms of Processing Time and Average Resource Utilization is presented in Fig. 21. The proposed method takes a processing time of 0.51 ms for 250 tasks, which is also lower compared to other methods, showing that it is efficient, meaning it responds quickly, which is beneficial for healthcare professionals in their decision-making processes. The visualization demonstrates that the proposed TP-DSDRL technique has a higher resource usage ratio than other techniques. When 600 tasks are involved, the TP-DSDRL technique has an impressive average resource usage of 84 %, which is higher than the other techniques. It shows that the proposed technique performs better in handling the task of offloading in the management of healthcare in the Cloud-Edge setup due to its efficiency.



(a)



(b)

Fig. 19. Comparison of proposed TP-DSDRL technique against existing techniques (a) Latency (b) Execution Time.

Fig. 22 illustrates the comparative analysis for various techniques under different task loads in terms of average task offloading delay and energy efficiency. The proposed TP-DSDRL method provides the lowest offloading delay of 21 ms when compared to other traditional works such as EA-MFO, DTH-ATB-MAPPO, DEEO, DPTARA, OO-DRL, DRL-LOA, and PBDLB. Meanwhile, from 250 tasks, the proposed TP-DSDRL technique has a high energy efficiency of 98.03 %, which surpasses that of other techniques. Increased energy efficiency results reveal that more tasks are performed with less energy consumption, especially for healthcare services that require continuous task execution with less battery consumption.

5.7. Ablation study

Table 3 depicts the ablation study for assessing the impact of key components in the proposed TP-DSDRL framework. When the HTCF method is eliminated, the capability in both the temporal dependency and contextual relationships capturing in healthcare signals are lost, resulting in low-accuracy extraction of features. The absence of HSI results in processing the healthcare tasks without contemplating the risk levels of the patient, thereby hindering the prioritization of important patient data. The elimination of the UAP-DSDRL method does not determine the decision of optimal task offloading between the resources of the device, edge, and cloud layers. For state processing, the basic neural representation is relied upon when the SCNN is eliminated, resulting in not efficient, non-informative, and compact representation learning. In the elimination of uncertainty-aware learning mechanism, the task offloading decisions are performed through the reinforcement learning agent without uncertainty consideration in resource availability and network conditions. The single objective is optimized by the reinforcement learning model when the multi objective rewarded function is removed. Furthermore, the simulation results indicate that the outcomes of the TP-DSDRL model dropped significantly without these components. On the other hand, when these components are incorporated, the system demonstrates exceptional outcomes. Overall, the study ensures that each component plays a vital role in achieving the high performance of the TP-DSDRL approach in healthcare task offloading.

5.8. Computational complexity analysis

Table 4 illustrates the computational complexity analysis of both the proposed TP-DSDRL approach and prior task offloading-based healthcare management strategies. Here, the proposed strategy acquired low

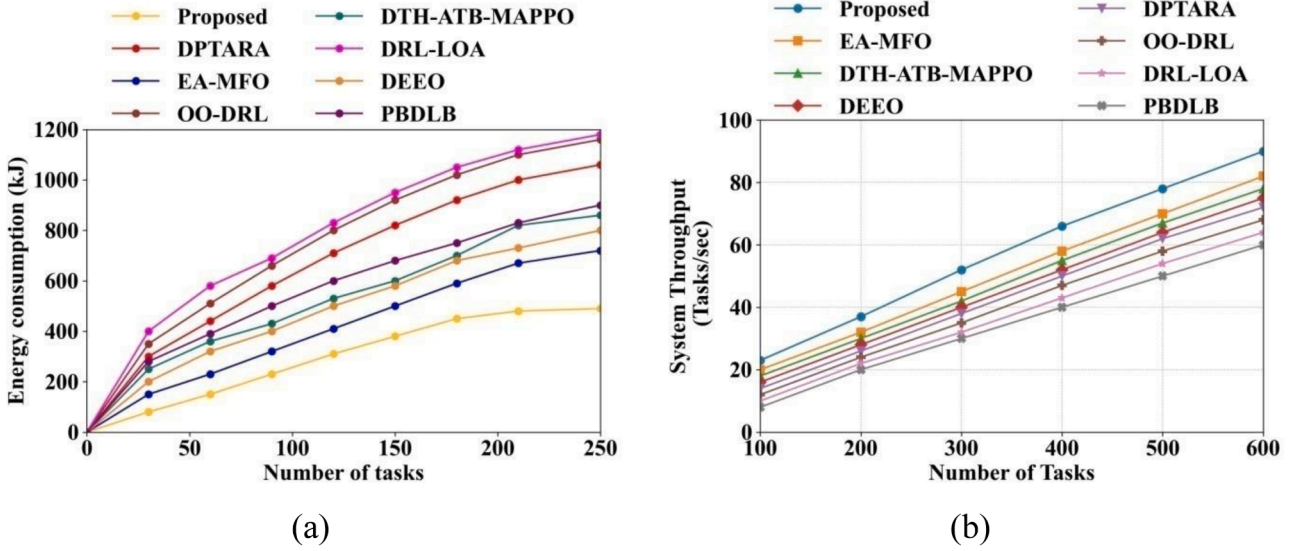


Fig. 20. Comparison of proposed TP-DSDRL technique against existing techniques (a) Energy Consumption (b) System Throughput.

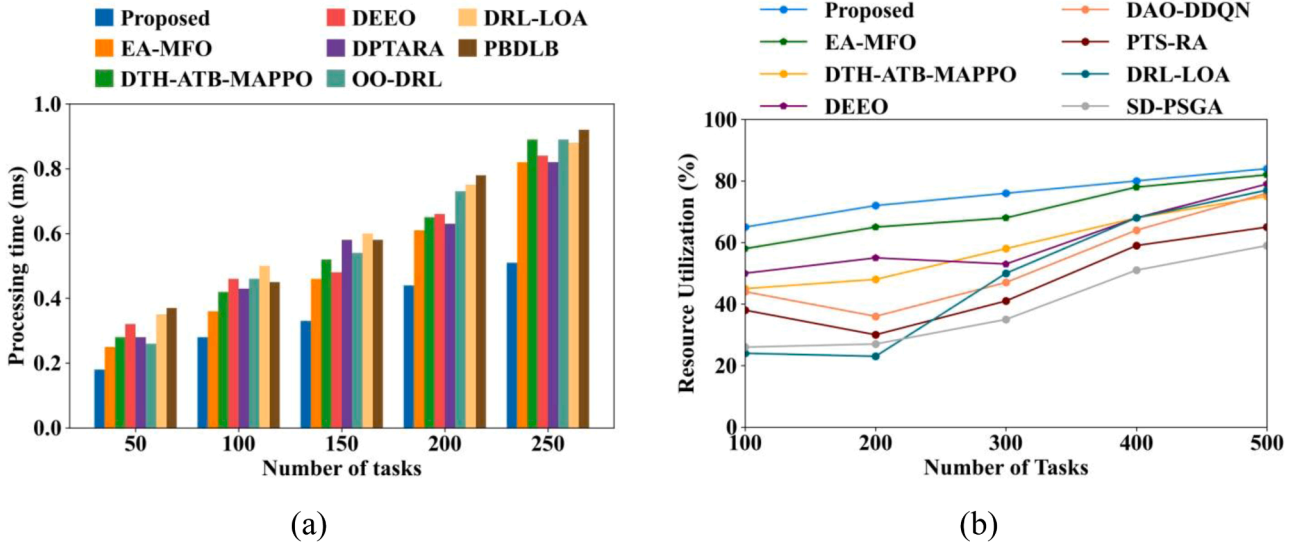


Fig. 21. Comparison of proposed TP-DSDRL technique against existing techniques (a) Processing Time (b) Average Resource Utilization.

computational complexity as compared to the EA-MFO, DTH-ATB-MAPPO, DEEO, DPTARA, OO-DRL, DRL-LOA, and PBDLB approaches. In addition, the pioneering mechanisms took more training time, but the proposed TP-DSDRL approach obtained a training time of 1.03 h. Table 4 ensures that the TP-DSDRL strategy not only minimizes computational complexity but also optimizes training and execution efficiency, which makes it a scalable solution for intelligent healthcare management systems.

6. Conclusion

This research proposed a novel TP-DSDRL strategy for intelligent task offloading in Cloud-Edge healthcare management systems. This framework comprises three hierarchical layers such as device, edge, and cloud layer, in which the physiological data are gathered in the device layer from IoT-enabled medical devices. In the edge layer, the gathered data are preprocessed through the preprocessing operations, including noise removal, normalization, and missing value handling, which are analyzed through the HTCF module by combining the transformer-based

attention and BiLSTM networks for extracting the features. The HSI is assessed for analyzing the risk level of the patients and allows dynamic task prioritization. A UAP-DSDRL method is implemented at the cloud layer for determining the best decision of task offloading. The SCNN is utilized for compact representation learning at the system state and Q values estimation for offloading actions. The decision reliability is improved through the integration of the uncertainty-aware learning mechanism and a multi-objective reward function. In this work, the efficiency of the proposed technique was tested by using two healthcare datasets. The experimental results showed that the proposed framework acquired higher performance outcomes, in which the proposed technique reduced the latency by 170 ms, the energy consumption of 490 kJ, Throughput of 90 Task/sec, Execution time of 620 ms, processing time of 0.51 ms, Average task offloading delay of 21 ms, energy efficiency of 98.03 % and Resource utilization of 84 %. The comparative analysis illustrated that the proposed technique outperformed previous methods regarding optimized task offloading in Cloud-Edge health care management systems. The TP-DSDRL method is extended in the future by integrating federated learning (FL) at the edge layer to allow the

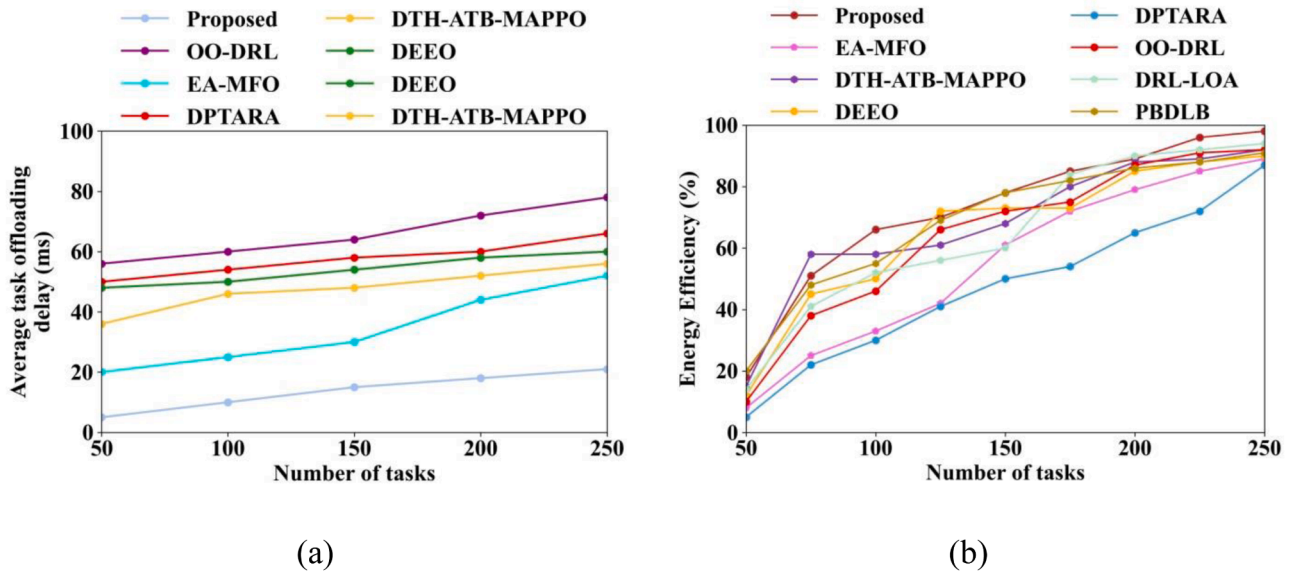


Fig. 22. Comparative analysis for various techniques under different task loads (a) Average Task Offloading Delay (b) Energy Efficiency.

Table 3

Ablation study for performance analysis of the proposed TP-DSDRL approach.

Techniques	Assessing Indicators							
	Latency (ms)	Execution time (ms)	Energy consumption (kJ)	System Throughput (Task/sec)	Processing time (ms)	Average resource utilization (%)	Average task offloading delay (ms)	Energy efficiency (%)
Proposed model	170	620	490	90	0.51	84	21	98.03
Without the HTCF method	210	685	570	82	0.68	88.92	39	92.82
Without HSI	250	700	590	79	0.71	90.93	42	90.65
Without the UAP-DSDRL method	253	715	610	76	0.73	91.05	44	90.01
Without SCNN	255	710	630	72	0.75	91.15	45	89.23
Without Uncertainty-Aware Learning Mechanism	259	716	660	61	0.79	93.21	47	86.92
Without a Multi-Objective Reward Function	270	725	710	59	0.83	95.65	51	85.33

Table 4

Computational Complexity Analysis of the proposed TP-DSDRL technique is compared with pioneering techniques.

Techniques	Number of parameters (M)	FLOPs (G)	Model size (MB)	Training time (hours)	Execution time (ms)
Proposed	1.09	17	34.14	1.03	620
EA-MFO	3.21	39.22	46.13	3.15	660
DTH-ATB-MAPPO	5.04	25	48	4.17	760
DEEO	3.75	47.10	40.12	3.47	700
DPTARA	3.11	36	53.76	3.58	850
OO-DRL	6.77	51.83	61.33	6.22	1100
DRL-LOA	3.59	34.90	55.98	4.10	1110
PBDLB	2.11	24.17	38.22	2.87	800

reinforcement learning model's collaborative training without sharing the sensitive data of the patient. The UAP-DSDRL policy and SCNN-based state representation of the proposed method are updated locally using edge nodes, which is resolved by the FL through synchronizing the parameters of the global model. The blockchain technology is incorporated along with the proposed method as the secure coordination layer for model record updating, participating node verification, and

preserving the tamper-resistant logs of data transactions and task offloading in healthcare environments.

Human and animal rights

This article does not contain any studies with human or animal subjects performed by any of the authors.

Informed consent

Not applicable.

CRediT authorship contribution statement

Kumaran K: Formal analysis, Data curation, Conceptualization. **Saranya G:** Methodology, Investigation, Funding acquisition. **Siva priya M S:** Writing – review & editing, Software, Resources, Project administration. **Vidhyalakshmi M:** Writing – original draft, Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- [1] N.S. Ganesh, V. Balasubramanian, D. Prasad, S.S. Velan, Deep learning-based user authentication with hybrid encryption for secured blockchain-aided data storage and optimal task offloading in mobile edge computing, *Wirel. Netw.* 31 (3) (2025) 2389–2417.
- [2] X. Gao, P. He, Y. Zhou, X. Qin, A smart healthcare system for remote areas based on the edge–cloud Continuum, *Electron. (Basel)* 13 (21) (2024) 4152.
- [3] X. Ji, F. Gong, N. Wang, J. Xu, X. Yan, Cloud-edge collaborative service architecture with large-tiny models based on deep reinforcement learning, *IEEE Trans. Cloud Comput.* 13 (2025) 288–302.
- [4] C. Sun, X. Li, C. Wang, Q. He, X. Wang, V.C. Leung, Hierarchical deep reinforcement learning for joint service caching and computation offloading in mobile edge-cloud computing, *IEEE Trans. Serv. Comput.* 17 (4) (2024) 1548–1564.
- [5] H. Min, A.M. Rahmani, P. Ghaderkourehpaz, K. Moghaddasi, M. Hosseinzadeh, A joint optimization of resource allocation management and multi-task offloading in high-mobility vehicular multi-access edge computing networks, *Ad. Hoc. Netw.* 166 (2025) 103656.
- [6] J. Li, M. Zhu, J. Liu, W. Liu, B. Huang, R. Liu, Blockchain-based reliable task offloading framework for edge-cloud cooperative workflows in IoMT, *Inf. Sci. (Ny)* 668 (2024) 120530.
- [7] B. Neha, S.K. Panda, P.K. Sahu, Task offloading and scheduling in osmotic computing-based health monitoring systems, in: 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), IEEE, 2024, pp. 1–6.
- [8] S.A.A. MS, M. DH, Task offloading and resource allocation using the echo tracking optimization enabled QoS-aware scheduling for MEC-enabled WBAN Healthcare system, *Trans. Emerg. Telecommun. Technol.* 36 (10) (2025) 70256.
- [9] W. Zhang, H. Ou, Reinforcement learning based multi objective task scheduling for energy efficient and cost effective cloud edge computing, *Sci. Rep.* 15 (1) (2025) 41716.
- [10] P. Consul, I. Budhiraja, R. Arora, S. Garg, B.J. Choi, M.S. Hossain, Federated reinforcement learning based task offloading approach for MEC-assisted WBAN-enabled IoMT, *Alex. Eng. J.* 86 (2024) 56–66.
- [11] B. Bhasker, S. Kaliraj, C. Gobinath, V. Sivakumar, Optimizing energy task offloading technique using IoMT cloud in healthcare applications, *J. Cloud Comput.* 14 (1) (2025) 9.
- [12] A.K. Jameil, H. Al-Raweshidy, Ai-enabled healthcare and enhanced computational resource management with digital twins into task offloading strategies, *IEEE Access* 12 (2024) 90353–90370.
- [13] A. Mondal, P.S. Chatterjee, N.K. Ray, An optimal novel approach for dynamic energy-efficient task offloading in mobile edge-cloud computing networks, *SN Comput. Sci.* 5 (5) (2024) 655.
- [14] J. Anand, B. Karthikeyan, Dynamic priority-based task scheduling and adaptive resource allocation algorithms for efficient edge computing in healthcare systems, *Results Eng.* 25 (2025) 104342.
- [15] X. Su, X. Fang, Z. Cheng, Z. Gong, C. Choi, Deep reinforcement learning based latency-energy minimization in smart healthcare network, *Digit. Commun. Netw.* 11 (3) (2025) 795–805.
- [16] Q. He, Z. Feng, Z. Chen, T. Nan, K. Li, H. Shen, K. Yu, X. Wang, Low-cost data offloading strategy with deep reinforcement learning for smart healthcare system, *IEEE Trans. Serv. Comput.* (2024).
- [17] S.S. Saranya, P. Anusha, S. Chandragandhi, O.K. Kishore, N.P. Kumar, K. Srihari, Enhanced decision-making in healthcare cloud-edge networks using deep reinforcement and lion optimization algorithm, *Biomed. Signal Process. Control* 92 (2024) 105963.
- [18] U. Shrawankar, S. Kokate, A. Jadhav, Optimized load balancing with task priority for reliable data management in IoT-fog-enabled healthcare systems, *Inf. Secur. J.: Glob. Perspect.* (2026) 1–17.
- [19] P. Choppa, B. Lokesh, Efficient task scheduling and load balancing in fog computing for crucial healthcare through deep reinforcement learning, *IEEE Access* 13 (2025) 26542–26563.
- [20] S. Khan, S. Liu, L. Pan, G. Mei, Optimization-based Hybrid offloading framework for IoMT in Edge-Cloud Healthcare systems, *Future Gener. Comput. Syst.* (2025) 108163.
- [21] S.M.H. Motlagh, M.H. Rezvani, M. Khounsiavash, AI methods for personality traits recognition: a systematic review, *Neurocomputing* 640 (2025) 130301.
- [22] S. Asif, Y. Wenhui, S. ur-Rehman, Q. ul-ain, K. Amjad, Y. Yueyang, S. Jinhai, M. Awais, Advancements and prospects of machine learning in medical diagnostics: unveiling the future of diagnostic precision, *Arch. Comput. Methods Eng.* 32 (2) (2025) 853–883.
- [23] Z. Chen, J. Zhang, G. Min, Z. Ning, J. Li, Traffic-aware lightweight hierarchical offloading toward adaptive slicing-enabled SAGIN, *IEEE J. Sel. Areas Commun.* 42 (12) (2024) 3536–3550.
- [24] Z. Chen, B. Xiong, Z. Chen, G. Min, J. Li, Joint computation offloading and resource allocation in multi-edge smart communities with personalized federated deep reinforcement learning, *IEEE Trans. Mob. Comput.* 23 (12) (2024) 11604–11619.
- [25] B. Xiao, C. Yu, X. Chen, Z. Chen, G. Min, Multi-agent collaboration for workflow task offloading in end-edge-cloud environments using deep reinforcement learning, *IEEE Trans. Parallel Distrib. Syst.* 36 (2025) 2281–2296.
- [26] D. Van Anh, V.H. Nguyen, Leveraging priority queueing in IoT-edge-fog-cloud infrastructures for efficient healthcare monitoring, *IEEE Access* 13 (2025) 80461–80477.
- [27] R. Greif, K.G. Lauridsen, T. Djärv, J.E. Ek, V. Monnelly, K.G. Monsieurs, N. Nikolaou, T.M. Olasveengen, F. Semeraro, A. Spartinou, J. Yeung, European Resuscitation Council guidelines 2025 executive summary, *Resuscitation* 215 (2025) 110770.
- [28] A. Thompson, K.E. Fleischmann, N.R. Smilowitz, L. de Las Fuentes, D. Mukherjee, N.R. Aggarwal, F.S. Ahmad, R.B. Allen, S.E. Altin, A. Auerbach, J.S. Berger, AHA/ACC/ACS/ASNC/HRS/SCA/SCCT/SCMR/SVM guideline for perioperative cardiovascular management for noncardiac surgery: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines, *J. Am. Coll. Cardiol.* 84 (19) (2024) 1869–1969.
- [29] M.A.S. Awan, B. Singla, A. Hasan, M.O.W. Abdalla, R. Altaf, S. Singla, S. Afaq, M. W. Mustafa, F. Asim, QTc interval prolongation in ICU patients: risk assessment and predictors, *Egypt. Heart J.* 78 (1) (2026) 5.